



CT126-3-2-CTF

PRACTICAL CTF STRATEGIES

GROUP: L-6 FLAG DEFENDER Z

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We would like to express our deepest gratitude to **Mr. Mohd Hanis Jenalis** for guiding us through every stage of this **Practical CTF Strategies** assignment. His clear explanations, quick support, and steady encouragement helped us crack each challenge with confidence. Thanks to his feedback, we understood the core concepts, sharpened our thinking, and learned how to break problems down the way a real CTF player should.

We are also grateful to our colleagues and peers. Their teamwork, idea sharing, and willingness to troubleshoot problems with us made every task easier to solve. A special thank you to our friends Sultan Abdulla Omar Takrori, Mohamed Abdifatah Ali, Ahmed Mirahusain Alvi, and Muhammad Taimoor Jamal for their constant support and motivation. They were always ready to help us decode tough moments and push forward.

This journey strengthened not just our technical skills but our problem-solving mindset, and we appreciate everyone who helped us level up along the way.

1. About the Competition

1.1 A Quick look at what the CTF Live Competition is all about

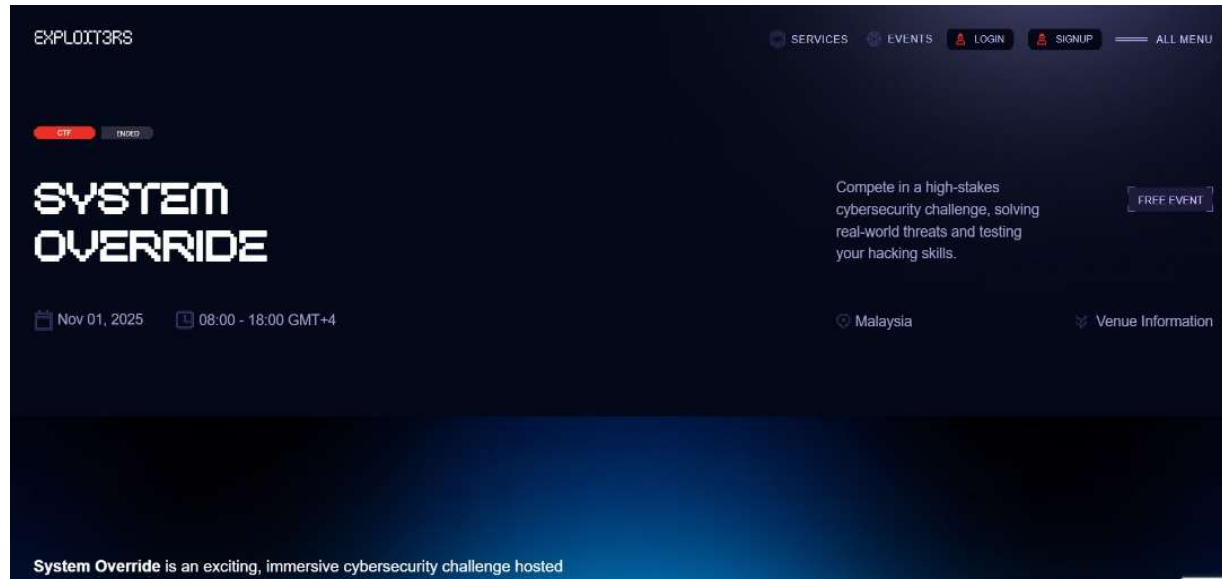


Figure 1. Homepage of System Override CTF Competition.

System Override 2025 is a Jeopardy-style Capture the Flag (CTF) competition hosted at Asia Pacific University (APU) and jointly organized by BarqSec and Exploit3rs, a professional cybersecurity company from United Arab Emirates (UAE). The CTF Competition was Held physically at Block S, Level 8, Room S-8-1 on 1 November 2025, the event ran from 10:00 AM to 5:30 PM and offered a wide range of challenges covering digital forensics, web exploitation, OSINT, networking, reverse engineering, and cryptography. The competition attracted over 300 participants and brought together teams from more than 10 universities across Malaysia, including UTP, UNITEN, UiTM, UKM, UPM, USIM, UTHM, and the German-Malaysian Institute, while also gaining support from returning APU alumni.

System Override follows the Jeopardy model, allowing teams to freely select challenges from multiple categories. Each challenge contains a hidden Flag that awards points when solved, helping teams climb the leaderboard. This structure promotes strategy, fast decision-making, and adaptability as participants shift between different cybersecurity domains under time pressure and stress that makes you forgot yourself (EXPLOIT3RS, 2025) .

Teams consisting of 1–5 members competed throughout the eight-hour window, and the venue was transformed into a cyber-themed battleground to create an immersive experience. Designed to support all skill levels, from beginner-friendly challenges to advanced exploitation tasks, making it suitable for newcomers and experienced competitors alike. Although the event featured bright neon visuals and an energetic atmosphere, once the challenge board unlocked, it quickly became a high-pressure technical sprint that demanded teamwork, problem-solving, and real cybersecurity skills. (*System Override 2025*).

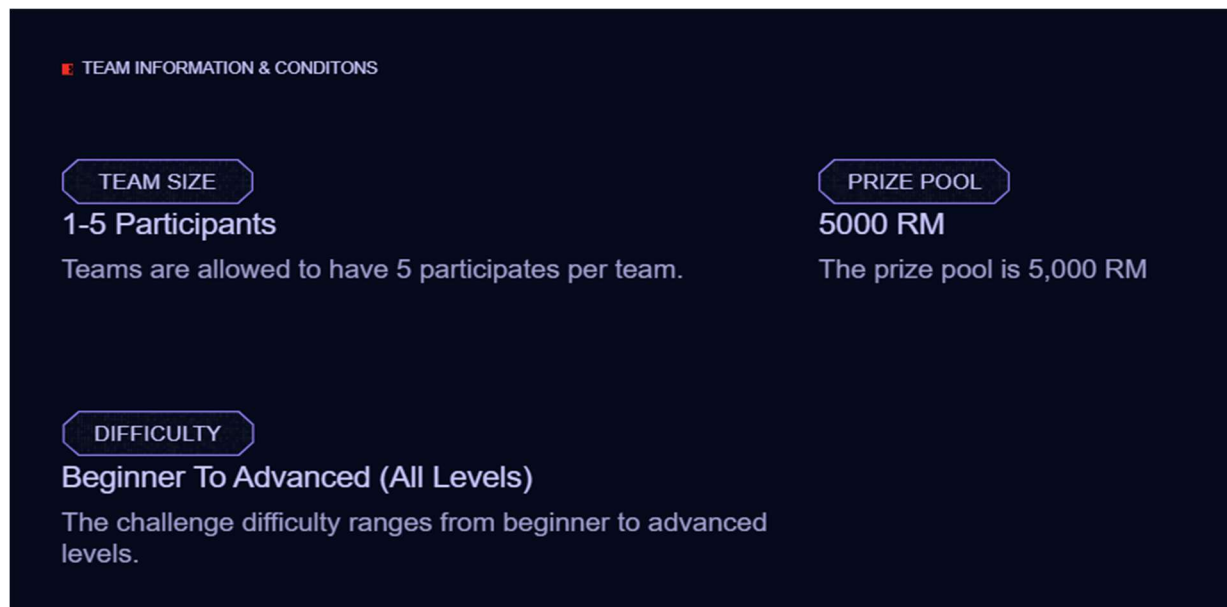


Figure 2. Competition Specifications: Team Size, Prize Pool, and Difficulty

System Override 2025 – Event Overview and Key Details:

- ❖ **Event Type:** Physical (On-Campus)
- ❖ **Format:** Jeopardy
- ❖ **Location:** Asia Pacific University (APU), Malaysia
- ❖ **Participants:** 300+ players
- ❖ **Universities:** 10+ universities across Malaysia
- ❖ **Teams:** Teams of 1–5 participants
- ❖ **Date & Time:** 1 November 2025 • 10:00 AM – 6:00 PM
- ❖ **Venue:** S-8-1, Level 8, Block S
- ❖ **Challenges:** Multiple categories including Web, Forensics, OSINT, Crypto, RE, and Networking

1.2 Prizes and Incentives



Figure 3. Cash Prize Breakdown for Top Three Winners.

System Override 2025 came with a solid **RM 5,000 prize pool**, which instantly boosted the competition energy. Only the Top 3 teams made it to the money, with the rewards split as:

- ❖ **1st Place:** RM 2,500
- ❖ **2nd Place:** RM 1,500
- ❖ **3rd Place:** RM 1,000

With prizes like that, the whole vibe shifted from “let’s just have fun” to “we’re not leaving without points.” Teams pushed harder, solved faster, argued less, and tried every trick they knew just to climb the leaderboard before the timer ended. (*System Override 2025*).

Even if a team didn’t make it into the Top 3, they still walked away with a **Certificate of Recognition**, proving that they showed up, competed, and survived the full eight-hour grind. So in the end, everyone left with something either cash, pride, or at least a certificate. The prize structure kept the competition intense but fun, making sure every player stayed locked in from start to finish.

1.3 Our Team - Arabseek

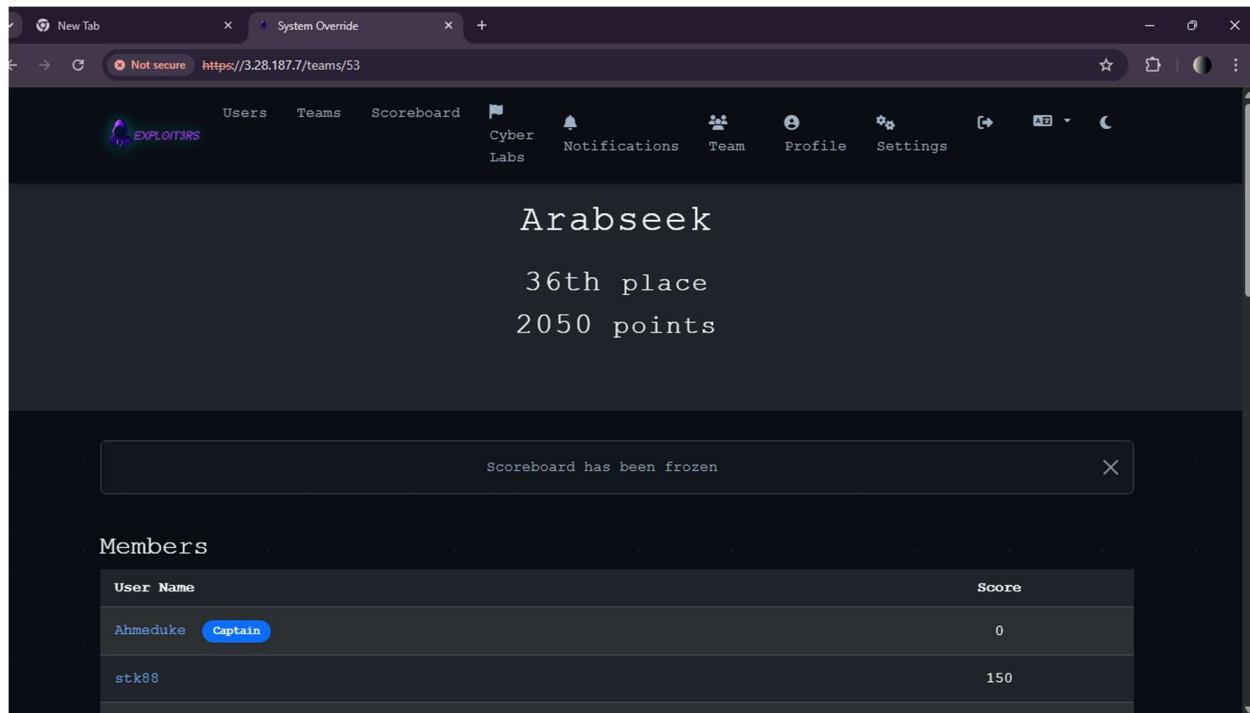


Figure 4. Team Arabseek Final Scoreboard Standing.

Arabseek is a team formed by Sultan Abdulla Takrori, together with Ahmed Alvi, Muhammad Taimoor Jamal, and Mohamed Abdifatah Ali. We decided to join System Override mainly because it caught our attention, and one of our friends recommended the event to us. Since we're students who enjoy learning through real challenges, the competition immediately felt like something we wanted to try.

For us, these events are more about gaining experience than chasing a ranking. Our interests naturally lean towards digital forensics, cryptography, and steganography, and these categories match our strengths and the way we think under pressure.

Communication was something that made our team work well. We understood each other's approach, so planning felt straightforward. Everyone knew their strengths, which helped us divide tasks quickly and avoid confusion. When one of us got stuck, another stepped in without hesitation. This kept our strategy simple, clear, and effective.

Our teamwork style allowed us to stay focused, support each other, and get the most out of every challenge we faced.

1.4 Received Official Authorization from our Lecturer

Before joining System Override 2025, we approached our lecturer, **Mr. Mohd Hanis Bin Jenalis**, to make sure the event aligned with our course learning outcomes. He encouraged us to participate after we explained our team's strengths in digital forensics, cryptography, and steganography, and he highlighted that this competition would be a strong learning opportunity. His approval gave us the confidence to treat the event not just as a challenge, but as a practical extension of what we study in class.

Since Mr. Hanis often promotes CTFs as a way to build real technical experience, his recommendation played a big role in our decision. System Override allowed us to apply concepts we've learned—such as investigation techniques, logical analysis, and data extraction—in a real CTF environment. It also pushed us to think critically under pressure while communicating as a team, which strengthened our problem-solving and coordination skills.

Instead of relying only on theory, this event helped us see how classroom knowledge translates into actual cybersecurity practice. For a team like ours, where every member has a different strength, this structured CTF format gave us the freedom to pick challenges that matched our skillsets and contribute equally. System Override ended up being a meaningful step in our growth, helping us bridge the gap between academic learning and real-world cybersecurity situations.

2. Challenge Description

2.1 Challenge Details

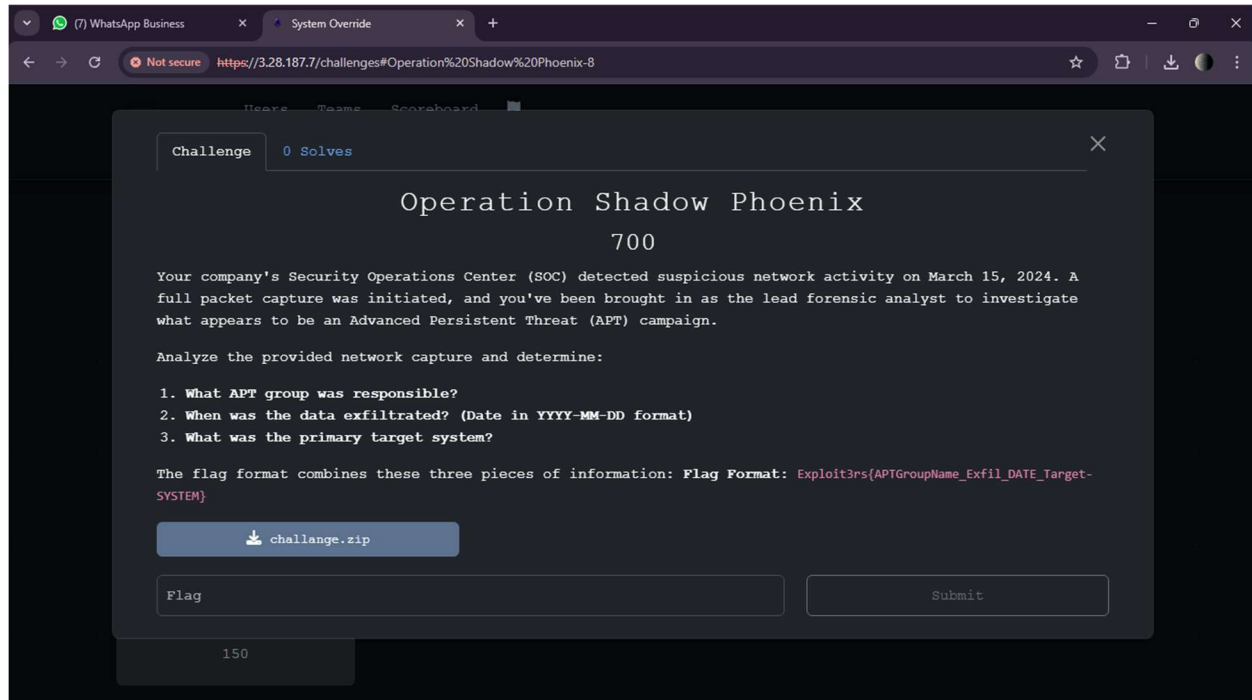


Figure 5. Details for the 'Operation Shadow Phoenix' Challenge.

Category: Networking

Challenge Name: Operation Shadow Phoenix

Difficulty Level: Medium-Hard (700 points)

Challenge Summary:

This challenge dropped us into a full-on SOC incident, where a suspicious network breach had already taken place. We were given a PCAP filled with hidden clues, weird traffic, and traces of an APT group trying to slip out with stolen data. Our task was to dig through the packets, follow every suspicious move, and uncover three critical details:

- ❖ The APT group responsible for the attack
- ❖ The exact date the data exfiltration occurred
- ❖ The primary system that was targeted.

It felt like a fast-paced investigation where every packet could reveal the next piece of the story.

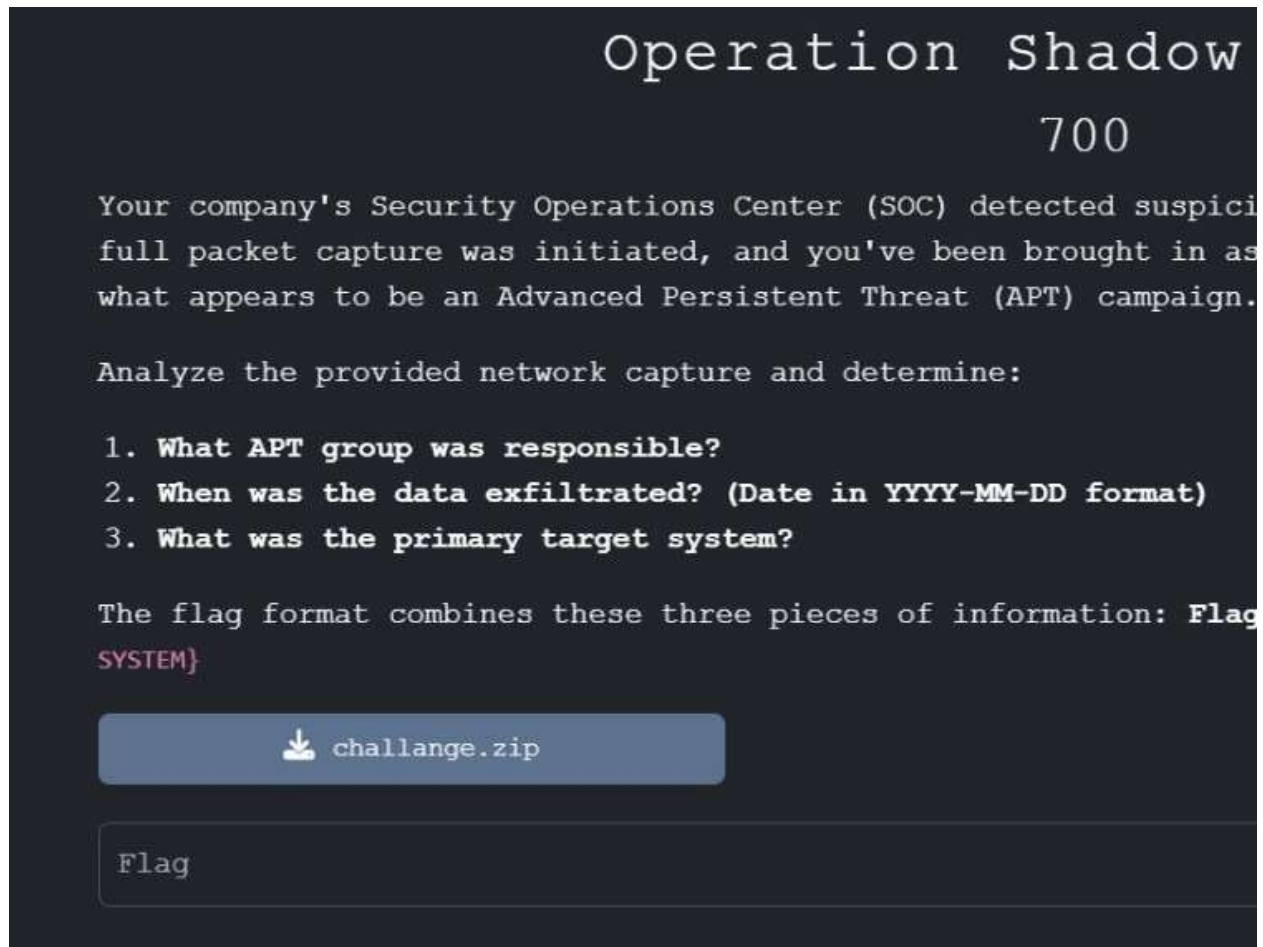


Figure 6. The Three Investigation Questions: APT Group, Exfiltration Date, and Target System.

The task required detailed network traffic analysis and was presented as a hard challenge, worth 700 points. Even though it looked straightforward at first, it demanded a solid understanding of Wireshark, especially following TCP/UDP streams, applying the right filters, and analyzing DNS traffic. We had to carefully read each part of the PCAP, check protocols like HTTP and SSH, and look for hidden or suspicious data. Because our team is comfortable with packet analysis and filtering, this challenge fit well with our skill set and allowed us to apply the techniques we have been learning in class.

2.2 Acquiring the Artifact

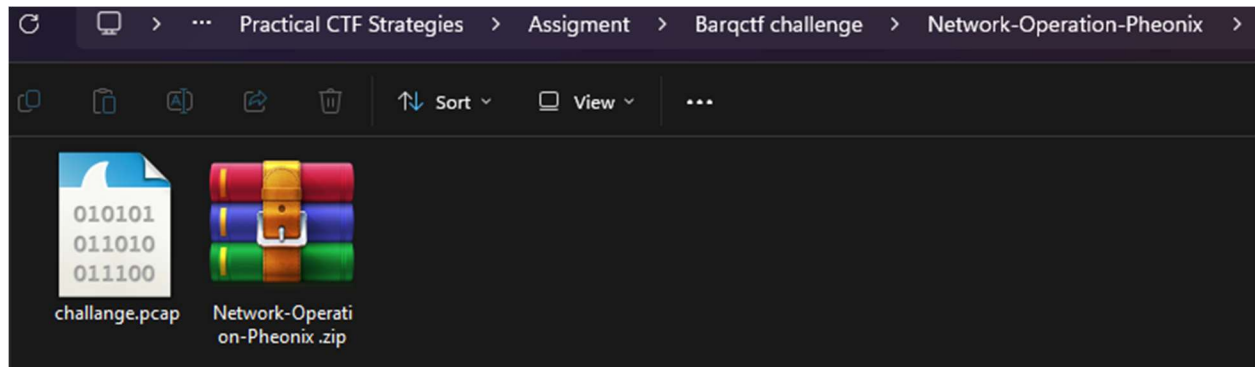


Figure 7. Directory View After Unzipping the Challenge Archive.

For this challenge, we were provided with a ZIP file containing the PCAP that held all the captured network traffic. After downloading and extracting it, we noticed that the PCAP file was fairly large, around **44.3 MB**, which already hinted that there would be a lot of packets to sort through and analyze. Since the entire investigation depended on understanding what happened inside this capture, our first step was to open the file using **Wireshark**, as it is the main tool for examining network packets and reconstructing what took place during the incident. Once loaded, we began exploring the traffic, following streams, applying filters, and preparing to trace the attacker's activity inside the network.

3. The Write-up

3.1 Initial Analysis of the Network Capture

Initial analysis of the PCAP file focused on identifying traffic patterns before performing detailed inspection. The capture revealed a mix of DNS and TCP packets (see Figure 8) from multiple internal sources, confirming that the dataset included significant background noise (legitimate traffic) in addition to the attacker's activity.

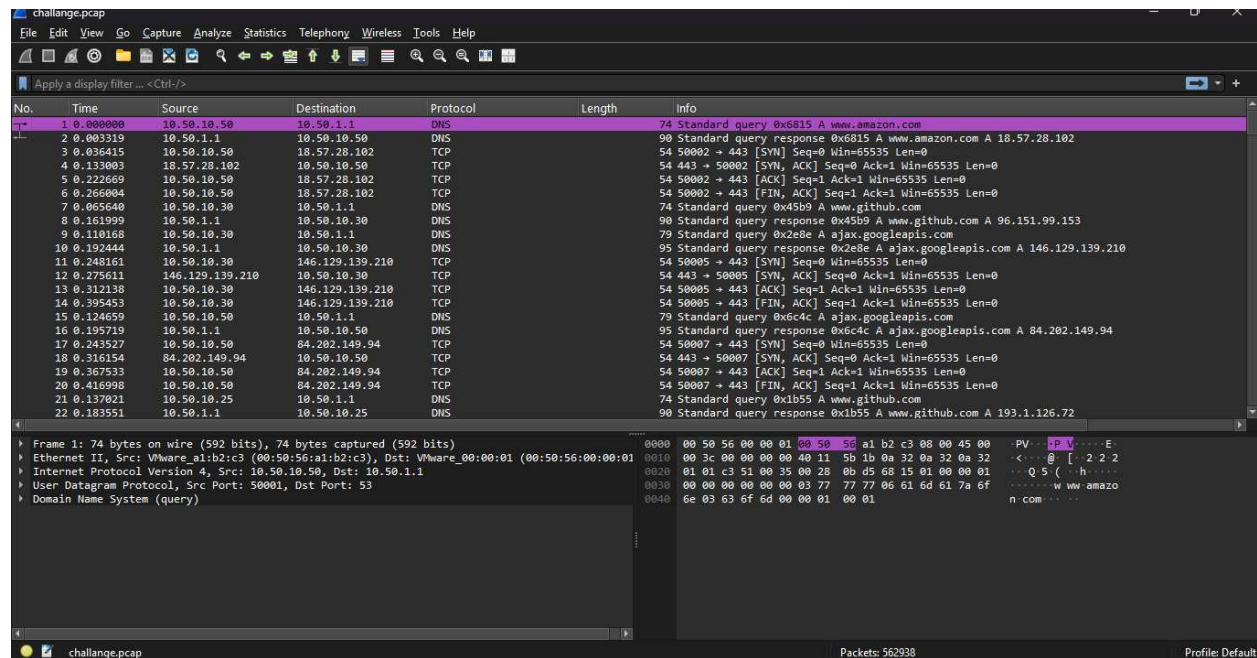


Figure 8. The 'challenge.pcap' File Loaded into Wireshark.

The objective of this phase was to characterize the general network traffic. We examined source and destination IP addresses and analyzed the protocol hierarchy to verify the active network layers. This preliminary profiling established a baseline of traffic density, enabling us to formulate effective filtration strategies for the subsequent deep packet inspection.

Initial inspection revealed a substantial dataset comprising 562,938 packets. Given this high volume of traffic, we determined that manual inspection was unfeasible. Consequently, our methodology relied heavily on Wireshark's advanced features, specifically Protocol Hierarchy statistics, TCP stream reconstruction, and display filters to effectively segregate malicious artifacts from the baseline legitimate traffic.

Subsequently, the analysis shifted toward granular traffic filtration and session reconstruction. The primary objective was to identify anomalous patterns and Indicators of Compromise (IoCs) that would facilitate the attribution of the APT group, establish the timeline of exfiltration, and pinpoint the compromised target system.

3.2 Analyzing the Protocol Hierarchy to Understand the Attacker's Strategy

Before applying specific filters, we sought to characterize the overall traffic composition of the PCAP file. We initiated this analysis by accessing the Protocol Hierarchy panel in Wireshark (see Figure 9), which provided a quantitative breakdown of the protocol distribution (see Figure 10).

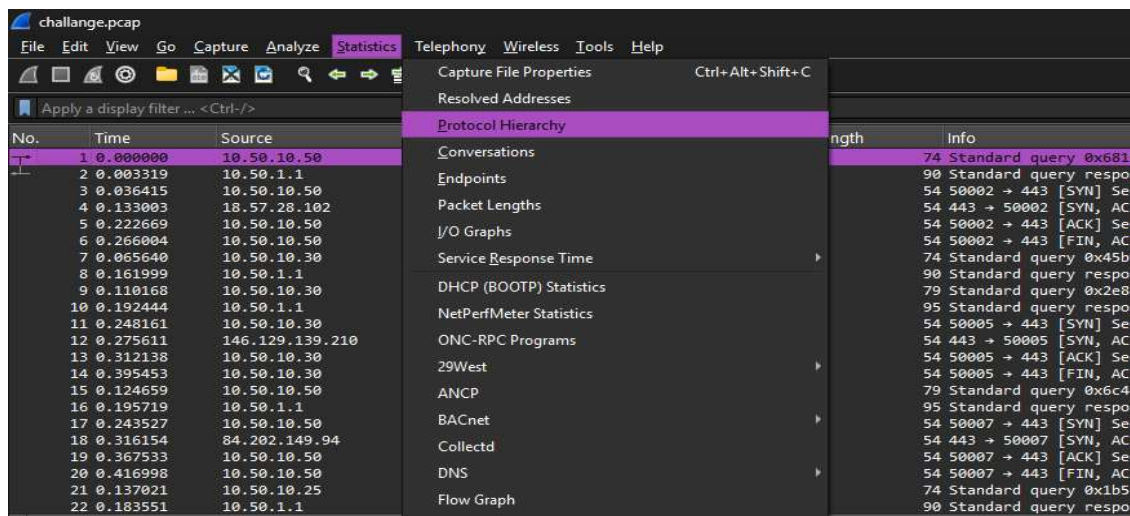


Figure 9. Accessing the Protocol Hierarchy Tool from the Main Menu.

Protocol	Percent Packets	Packets	Percent Bytes	Bytes	Bits/s	End Packets	End Bytes	End Bits/s	PDUs
Frame	100.0	562938	100.0	37547430	11 k	0	0	0	562938
Ethernet	100.0	562938	21.0	7881132	2493	0	0	0	562938
Internet Protocol Version 4	100.0	562938	30.0	11258760	3561	0	0	0	562938
User Datagram Protocol	41.6	234060	5.0	1872480	592	0	0	0	234060
Domain Name System	41.6	234060	26.4	9927366	3140	234060	9927366	3140	234060
Transmission Control Protocol	58.4	328878	17.5	6577560	2080	328848	6576960	2080	328878
Simple Mail Transfer Protocol	0.0	1	0.0	848	0	1	848	0	1
Data	0.0	29	0.0	1389	0	29	1389	0	29

Figure 10. Breakdown of Network Protocols and Packet Distribution.

Analysis of the Protocol Hierarchy revealed that the capture consisted exclusively of IPv4 traffic, split between UDP and TCP protocols. Crucially, the entirety of the UDP traffic was identified as DNS, which accounted for 41.6% of the total capture volume (see Figure 10). This abnormally

high concentration of domain resolution activity strongly suggests that the threat actor utilized DNS tunneling to obfuscate data exfiltration within this protocol.

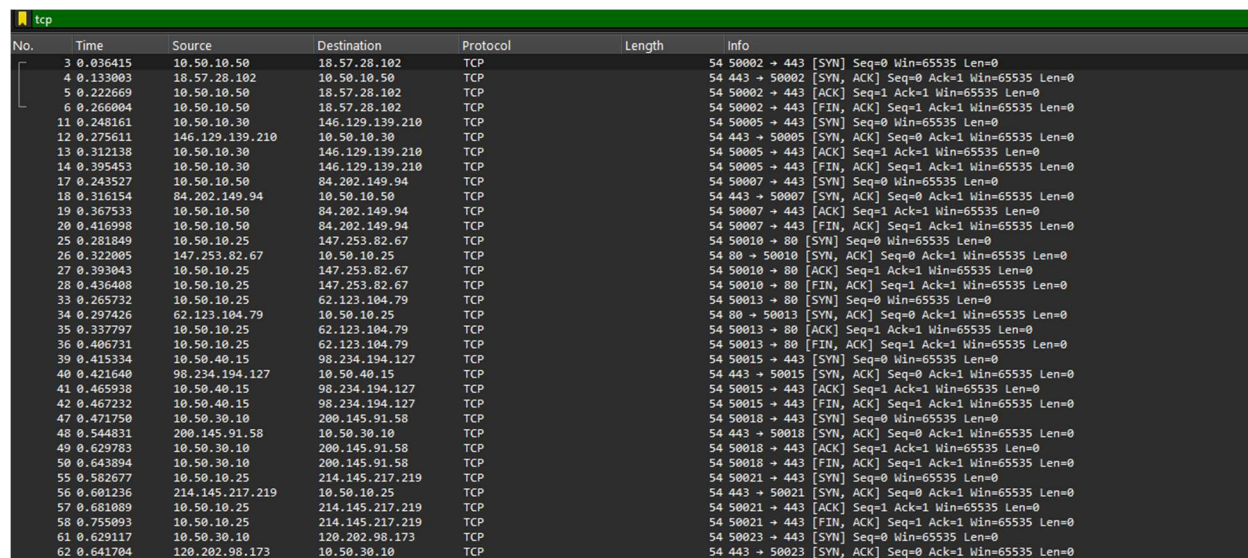
This analytical phase yielded two key insights:

1. **Traffic Characterization:** A comprehensive overview of the network traffic composition.
2. **Adversarial Methodology:** Preliminary indicators of the threat actor's tactics, techniques, and procedures (TTPs).

The evidence suggests a strategy of obfuscation, wherein the threat actor leveraged ubiquitous protocols to blend malicious activity with legitimate network behavior, thereby minimizing the likelihood of detection."

3.3 Applying filters and checking any traffic that Seems Suspicious

Following the initial characterization of the network traffic, we proceeded to apply display filters derived from the protocol hierarchy analysis. The investigation commenced with the isolation of TCP traffic to examine connection-oriented communications (see Figure 11).



No.	Time	Source	Destination	Protocol	Length	Info
3	0.036415	10.50.10.50	10.57.28.102	TCP	54	50002 → 443 [SYN] Seq=0 Win=65535 Len=0
4	0.133003	10.57.28.102	10.50.10.50	TCP	54	443 → 50002 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0
5	0.222669	10.50.10.50	10.57.28.102	TCP	54	50002 → 443 [ACK] Seq=1 Ack=1 Win=65535 Len=0
6	0.266004	10.50.10.50	10.57.28.102	TCP	54	50002 → 443 [FIN, ACK] Seq=1 Ack=1 Win=65535 Len=0
11	0.248161	10.50.10.30	146.129.139.210	TCP	54	50005 → 443 [SYN] Seq=0 Win=65535 Len=0
12	0.275611	146.129.139.210	10.50.10.30	TCP	54	443 → 50005 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0
13	0.312138	10.50.10.30	146.129.139.210	TCP	54	50005 → 443 [ACK] Seq=1 Ack=1 Win=65535 Len=0
14	0.395453	10.50.10.30	146.129.139.210	TCP	54	50005 → 443 [FIN, ACK] Seq=1 Ack=1 Win=65535 Len=0
17	0.243527	10.50.10.50	84.202.149.94	TCP	54	50007 → 443 [SYN] Seq=0 Win=65535 Len=0
18	0.316154	84.202.149.94	10.50.10.50	TCP	54	443 → 50007 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0
19	0.367533	10.50.10.50	84.202.149.94	TCP	54	50007 → 443 [ACK] Seq=1 Ack=1 Win=65535 Len=0
20	0.416998	10.50.10.50	84.202.149.94	TCP	54	50007 → 443 [FIN, ACK] Seq=1 Ack=1 Win=65535 Len=0
25	0.281849	10.50.10.25	147.253.82.67	TCP	54	50010 → 80 [SYN] Seq=0 Win=65535 Len=0
26	0.322005	147.253.82.67	10.50.10.25	TCP	54	80 → 50010 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0
27	0.393043	10.50.10.25	147.253.82.67	TCP	54	50010 → 80 [ACK] Seq=1 Ack=1 Win=65535 Len=0
28	0.436408	10.50.10.25	147.253.82.67	TCP	54	50010 → 80 [FIN, ACK] Seq=1 Ack=1 Win=65535 Len=0
33	0.265732	10.50.10.25	62.123.104.79	TCP	54	50013 → 80 [SYN] Seq=0 Win=65535 Len=0
34	0.297426	62.123.104.79	10.50.10.25	TCP	54	80 → 50013 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0
35	0.337797	10.50.10.25	62.123.104.79	TCP	54	50013 → 80 [ACK] Seq=1 Ack=1 Win=65535 Len=0
36	0.406731	10.50.10.25	62.123.104.79	TCP	54	50013 → 80 [FIN, ACK] Seq=1 Ack=1 Win=65535 Len=0
39	0.415334	10.50.40.15	98.234.194.127	TCP	54	50015 → 443 [SYN] Seq=0 Win=65535 Len=0
40	0.421640	98.234.194.127	10.50.40.15	TCP	54	443 → 50015 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0
41	0.465938	10.50.40.15	98.234.194.127	TCP	54	50015 → 443 [ACK] Seq=1 Ack=1 Win=65535 Len=0
42	0.467232	10.50.40.15	98.234.194.127	TCP	54	50015 → 443 [FIN, ACK] Seq=1 Ack=1 Win=65535 Len=0
47	0.471750	10.50.30.10	200.145.91.58	TCP	54	50018 → 443 [SYN] Seq=0 Win=65535 Len=0
48	0.544831	200.145.91.58	10.50.30.10	TCP	54	443 → 50018 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0
49	0.629783	10.50.30.10	200.145.91.58	TCP	54	50018 → 443 [ACK] Seq=1 Ack=1 Win=65535 Len=0
50	0.643894	10.50.30.10	200.145.91.58	TCP	54	50018 → 443 [FIN, ACK] Seq=1 Ack=1 Win=65535 Len=0
55	0.582677	10.50.10.25	214.145.217.219	TCP	54	50021 → 443 [SYN] Seq=0 Win=65535 Len=0
56	0.601236	214.145.217.219	10.50.10.25	TCP	54	443 → 50021 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0
57	0.681089	10.50.10.25	214.145.217.219	TCP	54	50021 → 443 [ACK] Seq=1 Ack=1 Win=65535 Len=0
58	0.755093	10.50.10.25	214.145.217.219	TCP	54	50021 → 443 [FIN, ACK] Seq=1 Ack=1 Win=65535 Len=0
61	0.629117	10.50.30.10	120.202.98.173	TCP	54	50023 → 443 [SYN] Seq=0 Win=65535 Len=0
62	0.641704	120.202.98.173	10.50.30.10	TCP	54	443 → 50023 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0

Figure 11. Network Traffic Filtered to Display Only TCP Packets.

Examination of the filtered TCP traffic revealed only standard connection establishment (three-way handshake) and termination sequences (SYN, SYN-ACK, ACK, followed by FIN/ACK), as illustrated in Figure 11. Analysis of the external traffic flows similarly indicated adherence to

standard protocol behaviors. Given that these patterns represent routine network connectivity, the traffic was classified as benign baseline activity and yielded no significant forensic value.

3.4 Finding Something Interesting in SMTP

Having determined that the general TCP traffic consisted of benign baseline activity, we shifted the investigative scope to the Simple Mail Transfer Protocol (SMTP).

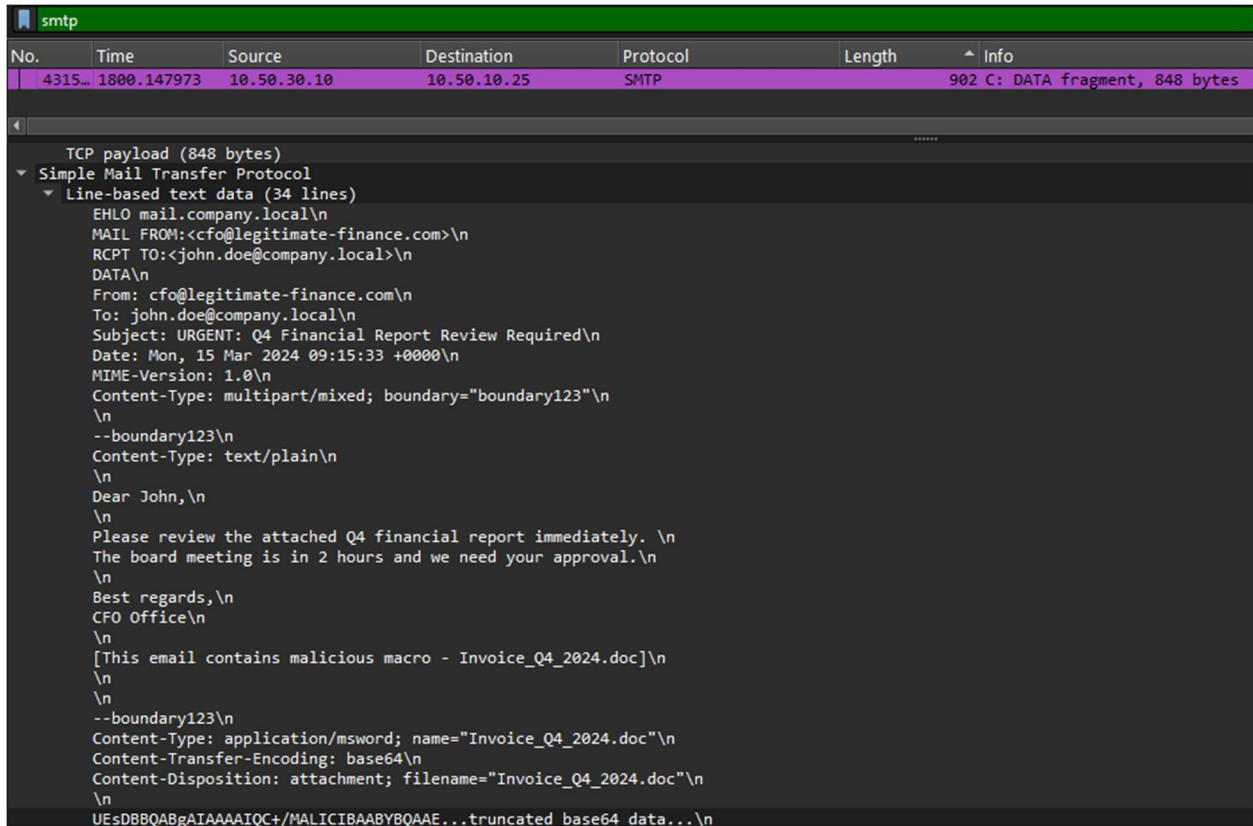


Figure 12. Filtered SMTP Traffic Revealing a Suspicious Email with Attachment.

Detailed inspection of the SMTP stream revealed a fully reconstructed email message, comprising standard headers, metadata, and a Base64-encoded attachment. The correspondence exhibited clear characteristics of a **social engineering attack**: it claimed to originate from the Chief Financial Officer (CFO) and leveraged urgency by requesting the review of a 'financial report.' Although the message body contained explicit references to a malicious macro, packet truncation within the PCAP file (see Figure 12, bottom pane) prevented the successful extraction and analysis of the attachment.

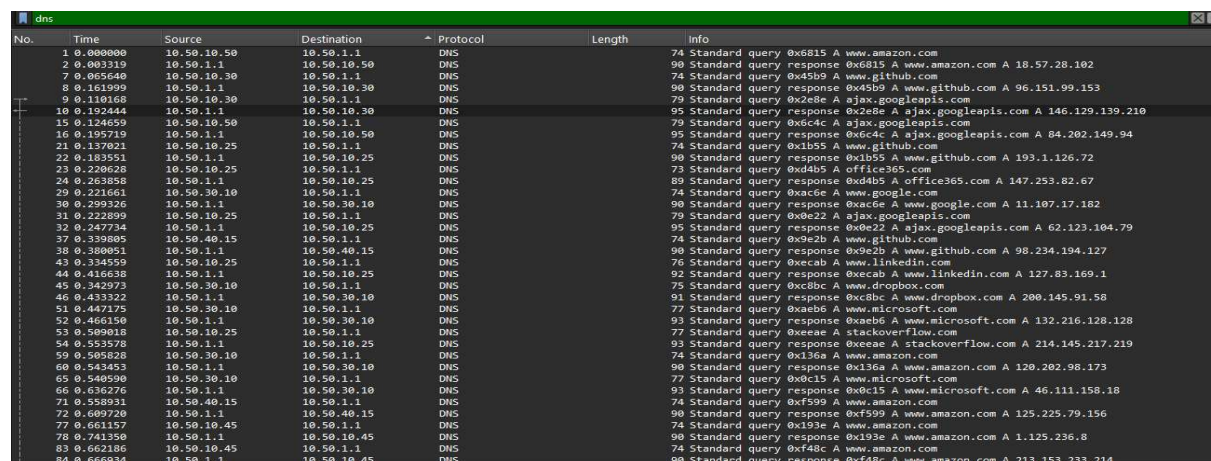
The analysis of this single packet provided critical intelligence regarding the adversary's methodology, establishing two key facts:

1. **Initial Access Vector:** The email served as the primary delivery mechanism, designed to breach the system perimeter.
2. **Payload Delivery:** The Base64-encoded attachment functioned as the malicious payload.

This evidence confirms that the attack lifecycle commenced with a targeted phishing campaign, which likely established the foothold necessary to facilitate the subsequent data exfiltration observed in the DNS traffic.

3.5 Moving to DNS to Hunt the Real Clues

Upon reviewing the DNS stream, we observed that a significant portion of the packets involved legitimate queries to reputable domains like www.amazon.com and www.github.com (see Figure 13, top right). These requests reflect typical user activity within a corporate environment and were identified as false positives for the purpose of this investigation.



No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	10.50.10.50	10.50.1.1	DNS	74	Standard query 0x6815 A www.amazon.com
2	0.003319	10.50.1.1	10.50.10.50	DNS	98	Standard query response 0x6815 A www.amazon.com A 18.57.28.102
7	0.005640	10.50.10.30	10.50.1.1	DNS	74	Standard query 0x45b9 A www.github.com
8	0.161999	10.50.1.1	10.50.10.30	DNS	98	Standard query response 0x45b9 A www.github.com A 96.151.99.153
9	0.110168	10.50.10.30	10.50.1.1	DNS	79	Standard query 0x2e8e A ajax.googleapis.com
10	0.192444	10.50.1.1	10.50.10.30	DNS	95	Standard query response 0x2e8e A ajax.googleapis.com A 146.129.139.210
15	0.124659	10.50.10.50	10.50.1.1	DNS	79	Standard query 0x6c4c A ajax.googleapis.com
16	0.195719	10.50.1.1	10.50.10.50	DNS	95	Standard query response 0x6c4c A ajax.googleapis.com A 84.202.149.94
21	0.137021	10.50.10.25	10.50.1.1	DNS	74	Standard query 0x1b55 A www.github.com
22	0.183551	10.50.1.1	10.50.10.25	DNS	98	Standard query response 0x1b55 A www.github.com A 193.1.126.72
23	0.220620	10.50.10.25	10.50.1.1	DNS	72	Standard query 0xd4b5 A office365.com
24	0.263858	10.50.1.1	10.50.10.25	DNS	89	Standard query response 0xd4b5 A office365.com A 147.253.82.67
29	0.221661	10.50.30.10	10.50.1.1	DNS	74	Standard query 0xac6e A www.google.com
30	0.299326	10.50.1.1	10.50.30.10	DNS	98	Standard query response 0xac6e A www.google.com A 11.107.17.182
31	0.222899	10.50.10.25	10.50.1.1	DNS	98	Standard query 0x9e2b A www.github.com
32	0.247734	10.50.1.1	10.50.10.25	DNS	95	Standard query response 0x9e2b A www.github.com A 62.123.104.79
37	0.339805	10.50.40.15	10.50.1.1	DNS	74	Standard query 0x9e2b A www.github.com
38	0.308051	10.50.1.1	10.50.40.15	DNS	98	Standard query response 0x9e2b A www.github.com A 98.234.194.127
43	0.334559	10.50.10.25	10.50.1.1	DNS	76	Standard query 0xecab A www.linkedin.com
44	0.416638	10.50.1.1	10.50.10.25	DNS	92	Standard query response 0xecab A www.linkedin.com A 127.83.169.1
45	0.342973	10.50.30.10	10.50.1.1	DNS	75	Standard query 0xcdbc A www.dropbox.com
46	0.433322	10.50.1.1	10.50.30.10	DNS	91	Standard query response 0xcdbc A www.dropbox.com A 200.145.91.58
51	0.447175	10.50.30.10	10.50.1.1	DNS	77	Standard query 0xae66 A www.microsoft.com
52	0.466150	10.50.1.1	10.50.30.10	DNS	93	Standard query response 0xae66 A www.microsoft.com A 132.216.128.128
53	0.509018	10.50.10.25	10.50.1.1	DNS	77	Standard query 0xeae A stackoverflow.com
54	0.553576	10.50.1.1	10.50.10.25	DNS	93	Standard query response 0xeae A stackoverflow.com A 214.145.217.219
59	0.505828	10.50.30.10	10.50.1.1	DNS	74	Standard query 0x136a A www.amazon.com
60	0.543453	10.50.1.1	10.50.30.10	DNS	98	Standard query response 0x136a A www.amazon.com A 120.202.98.173
65	0.540590	10.50.30.10	10.50.1.1	DNS	77	Standard query 0x0815 A www.microsoft.com
66	0.636276	10.50.1.1	10.50.30.10	DNS	93	Standard query response 0x0815 A www.microsoft.com A 46.111.158.18
71	0.558931	10.50.40.15	10.50.1.1	DNS	74	Standard query 0xf599 A www.amazon.com
72	0.609720	10.50.1.1	10.50.40.15	DNS	98	Standard query response 0xf599 A www.amazon.com A 125.225.79.156
77	0.651137	10.50.10.45	10.50.1.1	DNS	74	Standard query 0x193e A www.amazon.com
78	0.741350	10.50.1.1	10.50.10.45	DNS	98	Standard query response 0x193e A www.amazon.com A 1.125.236.8
83	0.662186	10.50.10.45	10.50.1.1	DNS	74	Standard query 0xf48c A www.amazon.com
84	0.666934	10.50.1.1	10.50.10.45	DNS	98	Standard query response 0xf48c A www.amazon.com A 213.153.233.214

Figure 13. Network Traffic Filtered to Display Only DNS Packets.

Initial visual inspection of the DNS traffic did not immediately reveal malicious artifacts. To isolate potential outliers, we sorted the dataset by packet length, operating on the hypothesis that data exfiltration payloads often result in packet sizes exceeding the standard baseline. This sorting strategy proved effective, revealing three specific DNS queries with significantly anomalous lengths compared to the remaining traffic (see Figure 14, bottom pane).

No.	Time	Source	Destination	Protocol	Length	Info
5627...	1797.803978	10.50.1.1	10.50.10.30	DNS	97	Standard query response 0x0dd2 A outlook.office365.com A 87.107.60.194
5629...	1799.674885	10.50.1.1	10.50.10.45	DNS	97	Standard query response 0x7e4c A outlook.office365.com A 137.224.108.151
5629...	1799.751936	10.50.1.1	10.50.30.10	DNS	97	Standard query response 0xfale A outlook.office365.com A 119.194.240.69
4317...	3600.035485	10.50.1.1	10.50.10.25	DNS	102	Standard query response 0xf510 A cdn-updates.azureedge.info A 185.220.101.45
4316...	4200.043534	10.50.1.1	10.50.10.25	DNS	102	Standard query response 0x14dc A cdn-updates.azureedge.info A 185.220.101.45
4316...	5400.099329	10.50.1.1	10.50.10.25	DNS	102	Standard query response 0x4ab9 A cdn-updates.azureedge.info A 185.220.101.45
4316...	5700.106438	10.50.1.1	10.50.10.25	DNS	102	Standard query response 0xbeld A cdn-updates.azureedge.info A 185.220.101.45
4316...	6600.068959	10.50.1.1	10.50.10.25	DNS	102	Standard query response 0x1ae9 A cdn-updates.azureedge.info A 185.220.101.45
4316...	6900.002969	10.50.1.1	10.50.10.25	DNS	102	Standard query response 0x3b71 A cdn-updates.azureedge.info A 185.220.101.45
4316...	7200.023573	10.50.1.1	10.50.10.25	DNS	102	Standard query response 0xeb4d A cdn-updates.azureedge.info A 185.220.101.45
4316...	7799.985936	10.50.1.1	10.50.10.25	DNS	102	Standard query response 0x8028 A cdn-updates.azureedge.info A 185.220.101.45
4317...	8700.041472	10.50.1.1	10.50.10.25	DNS	102	Standard query response 0x9833 A cdn-updates.azureedge.info A 185.220.101.45
4317...	9000.036262	10.50.1.1	10.50.10.25	DNS	102	Standard query response 0x6f48 A cdn-updates.azureedge.info A 185.220.101.45
4317...	9300.044522	10.50.1.1	10.50.10.25	DNS	102	Standard query response 0x2000 A cdn-updates.azureedge.info A 185.220.101.45
4317...	10200.046913	10.50.1.1	10.50.10.25	DNS	102	Standard query response 0x545f A cdn-updates.azureedge.info A 185.220.101.45
4317...	10500.074710	10.50.1.1	10.50.10.25	DNS	102	Standard query response 0xcc08 A cdn-updates.azureedge.info A 185.220.101.45
4317...	11100.093235	10.50.1.1	10.50.10.25	DNS	102	Standard query response 0x832b A cdn-updates.azureedge.info A 185.220.101.45
4316...	4500.005502	10.50.1.1	10.50.10.25	DNS	108	Standard query response 0x701f A api-telemetry.cloudflare-cdn.net A 185.220.101.45
4316...	5100.005800	10.50.1.1	10.50.10.25	DNS	108	Standard query response 0x6004 A api-telemetry.cloudflare-cdn.net A 185.220.101.45
4317...	8399.989331	10.50.1.1	10.50.10.25	DNS	108	Standard query response 0xdb5d A api-telemetry.cloudflare-cdn.net A 185.220.101.45
4317...	9600.005523	10.50.1.1	10.50.10.25	DNS	108	Standard query response 0x6188 A api-telemetry.cloudflare-cdn.net A 185.220.101.45
4317...	10800.009804	10.50.1.1	10.50.10.25	DNS	108	Standard query response 0x49db A api-telemetry.cloudflare-cdn.net A 185.220.101.45
4316...	4800.006424	10.50.1.1	10.50.10.25	DNS	109	Standard query response 0x9a08 A metrics-analytics.googleapis.info A 185.220.101.45
4316...	6800.078189	10.50.1.1	10.50.10.25	DNS	109	Standard query response 0x53a9 A metrics-analytics.googleapis.info A 185.220.101.45
4316...	6300.046722	10.50.1.1	10.50.10.25	DNS	109	Standard query response 0x442c A metrics-analytics.googleapis.info A 185.220.101.45
4316...	7500.101885	10.50.1.1	10.50.10.25	DNS	109	Standard query response 0x3426 A metrics-analytics.googleapis.info A 185.220.101.45
4317...	8100.109420	10.50.1.1	10.50.10.25	DNS	109	Standard query response 0xacce A metrics-analytics.googleapis.info A 185.220.101.45
4317...	9900.008122	10.50.1.1	10.50.10.25	DNS	109	Standard query response 0x28aa A metrics-analytics.googleapis.info A 185.220.101.45
4331...	21600.031944	10.50.1.1	10.50.10.25	DNS	112	Standard query response 0xb084 A data-shadowphoenix.cloudservices.net A 190.51.100.88
4333...	21934.167639	10.50.10.25	10.50.1.1	DNS	141	Standard query 0x7c35 A exfil-00-03-553268685a473933554768765a573570.data-shadowphoenix.
4333...	21936.769467	10.50.10.25	10.50.1.1	DNS	141	Standard query 0x8961 A exfil-01-03-65487794d4449304c544174c544531.data-shadowphoenix.
4333...	21938.282237	10.50.10.25	10.50.1.1	DNS	141	Standard query 0x3975 A exfil-02-03-66455a70626d4675932564551673d3d.data-shadowphoenix.
4333...	21934.178411	10.50.1.1	10.50.10.25	DNS	157	Standard query response 0x7c35 A exfil-00-03-553268685a473933554768765a573570.data-shado
4333...	21936.864399	10.50.1.1	10.50.10.25	DNS	157	Standard query response 0x8961 A exfil-01-03-65487794d4449304c544174c544531.data-shado
4333...	21934.178411	10.50.1.1	10.50.10.25	DNS	157	Standard query response 0x7c35 A exfil-00-03-553268685a473933554768765a573570.data-shado
4333...	21938.282237	10.50.10.25	10.50.1.1	DNS	141	Standard query 0x3975 A exfil-02-03-66455a70626d4675932564551673d3d.data-shadowphoenix.
4333...	21936.769467	10.50.10.25	10.50.1.1	DNS	141	Standard query 0x8961 A exfil-01-03-65487794d4449304c544174c544531.data-shado
4333...	21934.167639	10.50.10.25	10.50.1.1	DNS	141	Standard query 0x7c35 A exfil-00-03-553268685a473933554768765a573570.data-shado
4331...	21600.031944	10.50.1.1	10.50.10.25	DNS	112	Standard query response 0xb084 A data-shado

Figure 14. DNS Traffic Filtered and Sorted by Packet Length.

We distinguished this traffic from normal operations due to the anomalous length of the domain names. Analysis of the packet details revealed that the domains consisted of long, high-entropy strings, suggesting the use of DNS tunneling to conceal data exfiltration (see Figure 15).

No.	Time	Source	Destination	Protocol	Length	Info
4333...	21938.316886	10.50.1.1	10.50.10.25	DNS	157	Standard query response 0x3975 A exfil-02-03-66455a70626d4675932564551673d3d.data-shadowphoenix.
4333...	21936.864399	10.50.1.1	10.50.10.25	DNS	157	Standard query response 0x8961 A exfil-01-03-65487794d4449304c544174c544531.data-shadowphoenix.
4333...	21934.178411	10.50.1.1	10.50.10.25	DNS	157	Standard query response 0x7c35 A exfil-00-03-553268685a473933554768765a573570.data-shadowphoenix.
4333...	21938.282237	10.50.10.25	10.50.1.1	DNS	141	Standard query 0x3975 A exfil-02-03-66455a70626d4675932564551673d3d.data-shadowphoenix.
4333...	21936.769467	10.50.10.25	10.50.1.1	DNS	141	Standard query 0x8961 A exfil-01-03-65487794d4449304c544174c544531.data-shadowphoenix.
4333...	21934.167639	10.50.10.25	10.50.1.1	DNS	141	Standard query 0x7c35 A exfil-00-03-553268685a473933554768765a573570.data-shadowphoenix.

Frame 433365: 141 bytes on wire (1128 bits), 141 bytes captured (1128 bits)

Ethernet II, Src: VMware_a1:b2:c3 (00:50:56:a1:b2:c3), Dst: VMware_00:00:01 (00:50:56:00:00:01)

Internet Protocol Version 4, Src: 10.50.10.25, Dst: 10.50.1.1

User Datagram Protocol, Src Port: 63421, Dst Port: 53

Domain Name System (query)

Transaction ID: 0x7c35

Flags: 0x0100 Standard query

Questions: 1

Answer RRs: 0

Authority RRs: 0

Additional RRs: 0

Queries

exfil-00-03-553268685a473933554768765a573570.data-shadowphoenix.cloudservices.net: type A, class IN

[Response in: 433366]

Figure 15. Encoded String Identified in Suspicious High-Length DNS Queries.

We further identified three matching response packets (see Figure 15, top right pane) verifying that the external server received and acknowledged the suspicious queries. Although these packets utilized valid DNS headers, their anomalous payload structure and length provided definitive evidence that they were not standard lookup operations.

3.6 Investigating the Suspicious DNS Queries Further

We isolated the three suspect DNS queries previously identified by their excessive length and irregular structure. To prepare for analysis, the obfuscated strings contained within the domain name fields were extracted directly from the packet payload (see Figure 16).

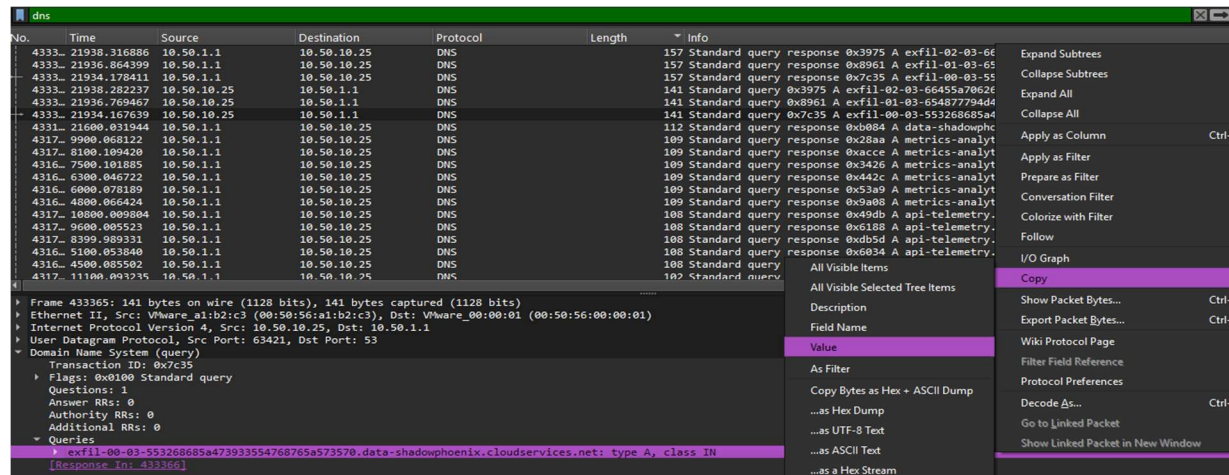


Figure 16. Extracting the Encoded String Using the 'Copy Value' Option.

At this stage, the objective was not to perform immediate decoding, but rather to isolate the specific character strings to facilitate a structured analysis.

3.7 Asking ChatGPT for Initial Interpretation

Following the extraction of the anomalous domain strings, we utilized a Large Language Model (see Figure 17) to expedite the identification of the encoding scheme.

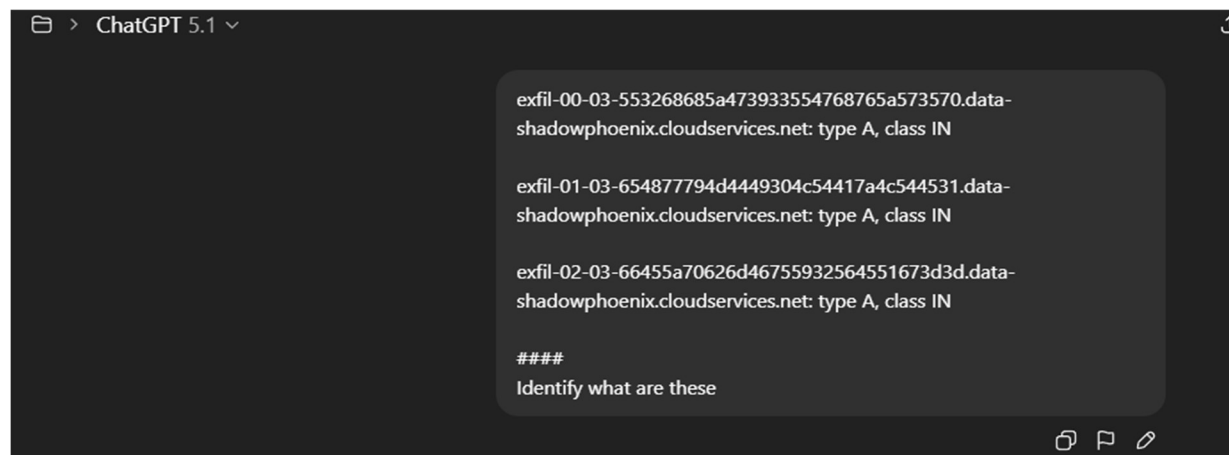


Figure 17. Querying ChatGPT (an LLM) to Analyze the Extracted DNS Data Payload.

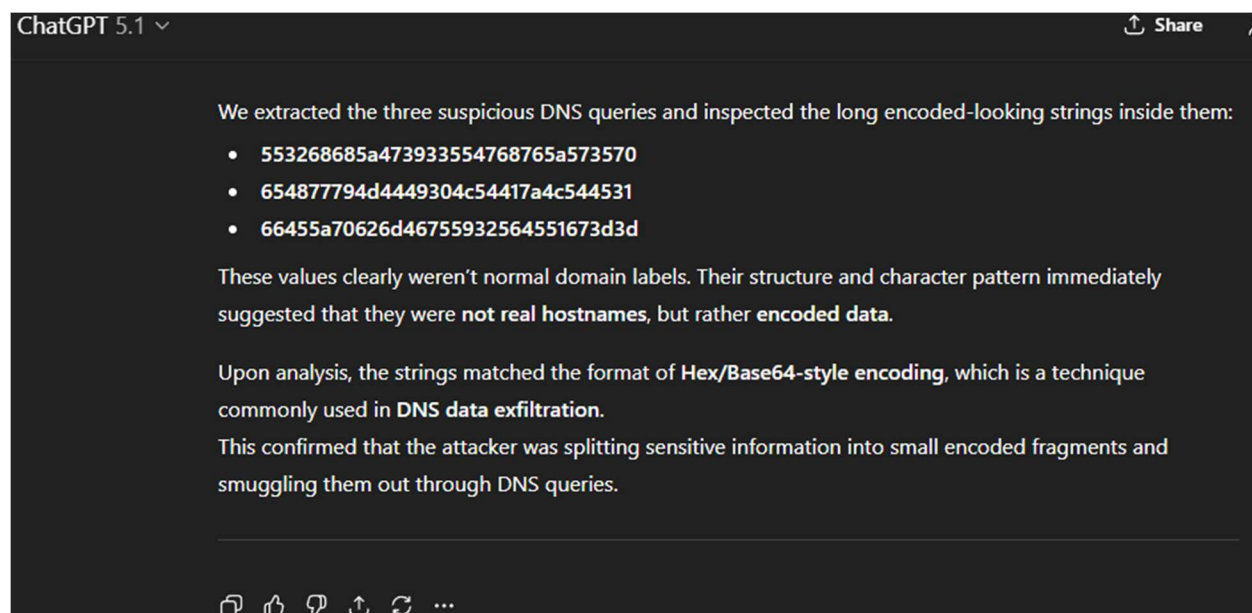


Figure 18. ChatGPT Analysis Confirming Hex/Base64 style Encoding and DNS Exfiltration.

Consultation with the Large Language Model (ChatGPT) validated that the subdomain strings deviated from standard domain nomenclature. The analysis identified the character patterns as consistent with Hexadecimal (Base16) encoding, a technique frequently employed by threat actors for payload obfuscation. This finding substantiated the hypothesis that the DNS protocol was being utilized as a covert channel for data exfiltration.

3.8 Decrypting the Exfiltrated Strings using CyberChef

The decoding process commenced with the analysis of the initial extracted artifact:

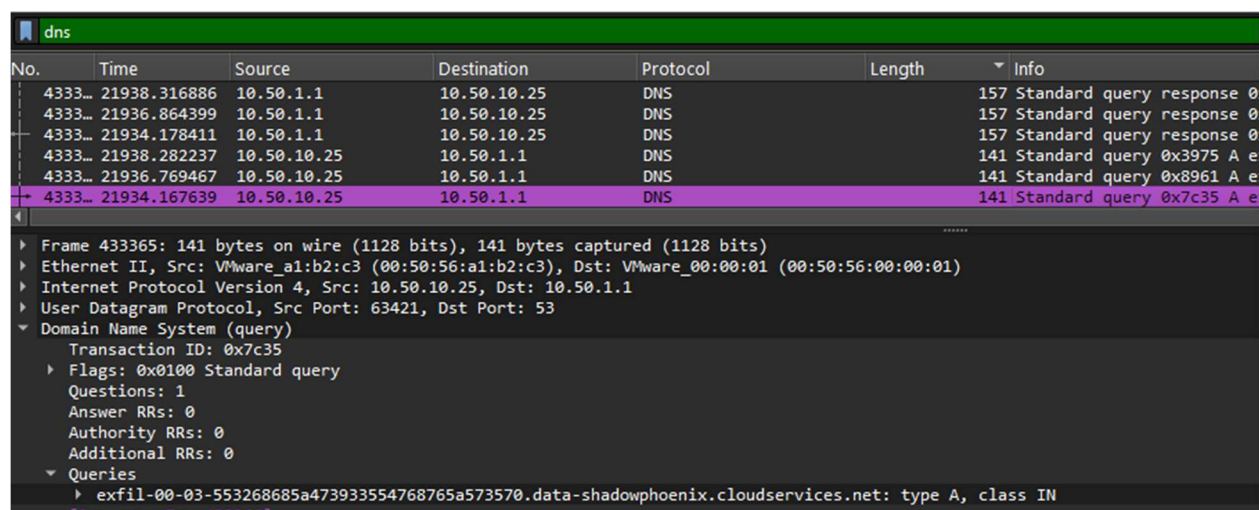


Figure 19. Selection of the First Encoded Hex String for Decryption.

Upon extracting the hexadecimal payload, the artifact was imported into the CyberChef analysis framework. The decoding process was executed by applying the 'From Hex' operation to the input string.

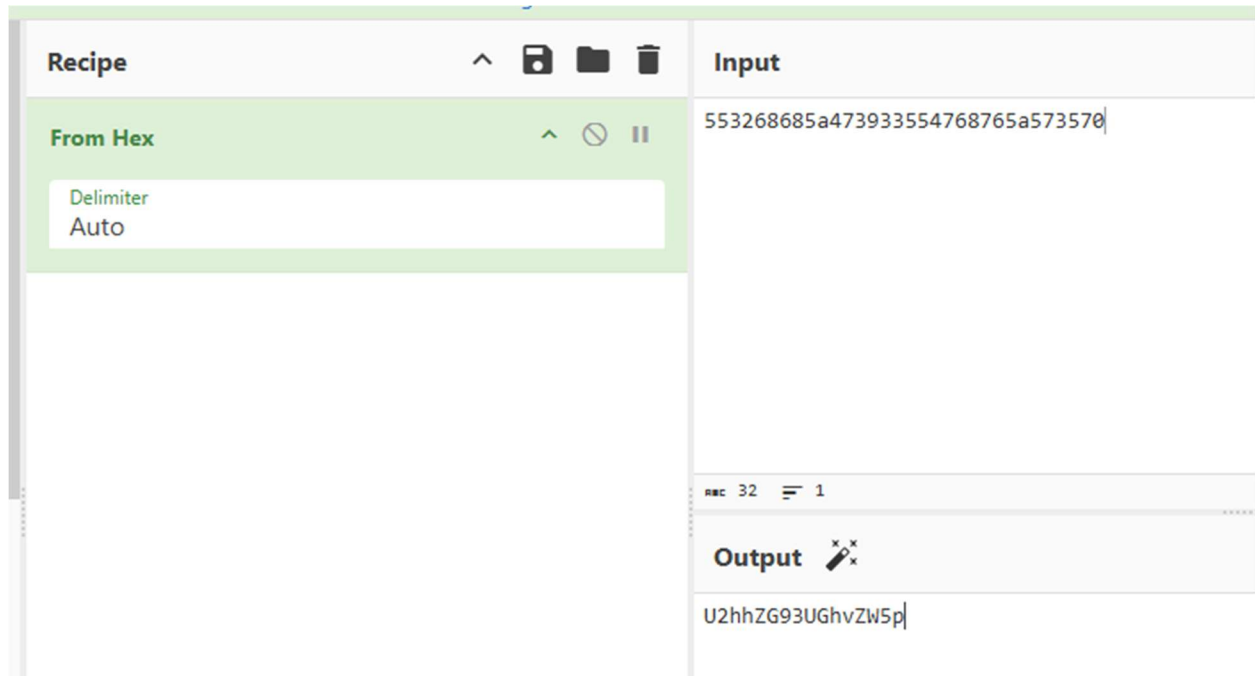


Figure 20. CyberChef Output Revealing a Second Layer of Encoding.

The initial hexadecimal conversion yielded an intermediate string that remained obfuscated (see Figure 20). However, visual inspection of the character set specifically the combination of uppercase and lowercase alphanumeric characters strongly indicated a secondary layer of Base64 encoding. Consequently, we applied the 'From Base64' operation to the intermediate payload.

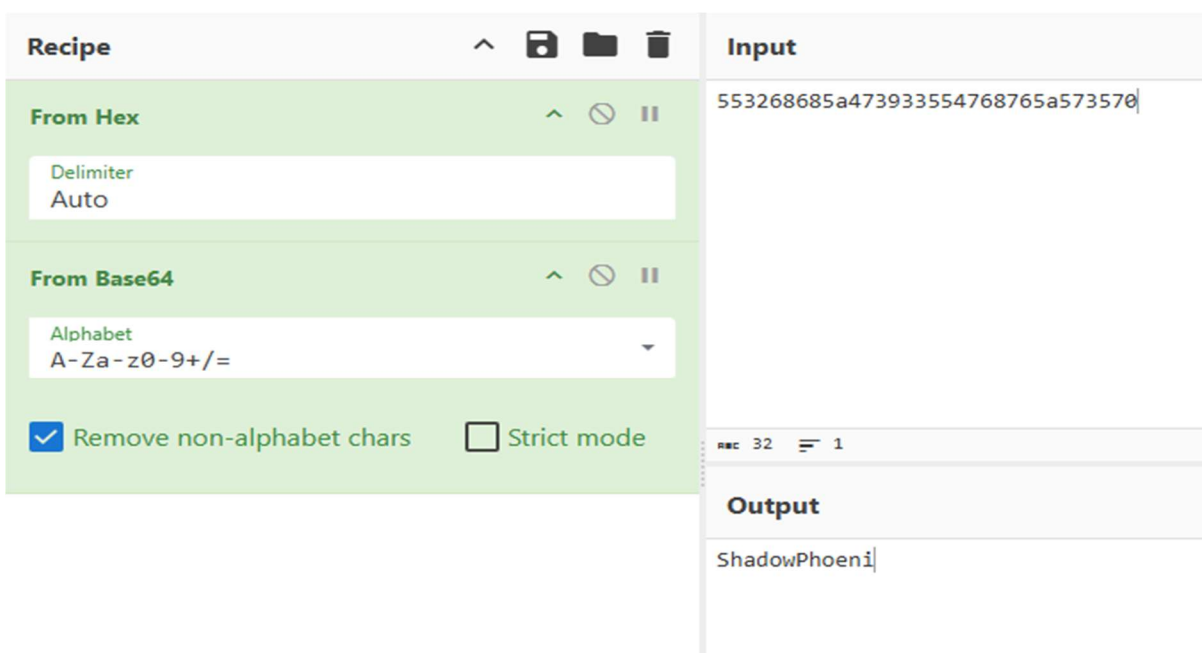


Figure 21. Final Base64 Decoding to Reveal the APT Group Responsible for the Attack.

The execution of this secondary decoding operation immediately yielded the legible plaintext string: 'Shadow Phoenix ' (see Figure 21). This artifact provided critical attribution intelligence, identifying the threat actor as the '**Shadow Phoenix**' APT group. The successful recovery of this string confirmed that the adversary employed a double-encoding scheme (Hexadecimal followed by Base64) to obfuscate the payload within the DNS queries.

Subsequently, we targeted the second obfuscated payload for extraction and decoding:

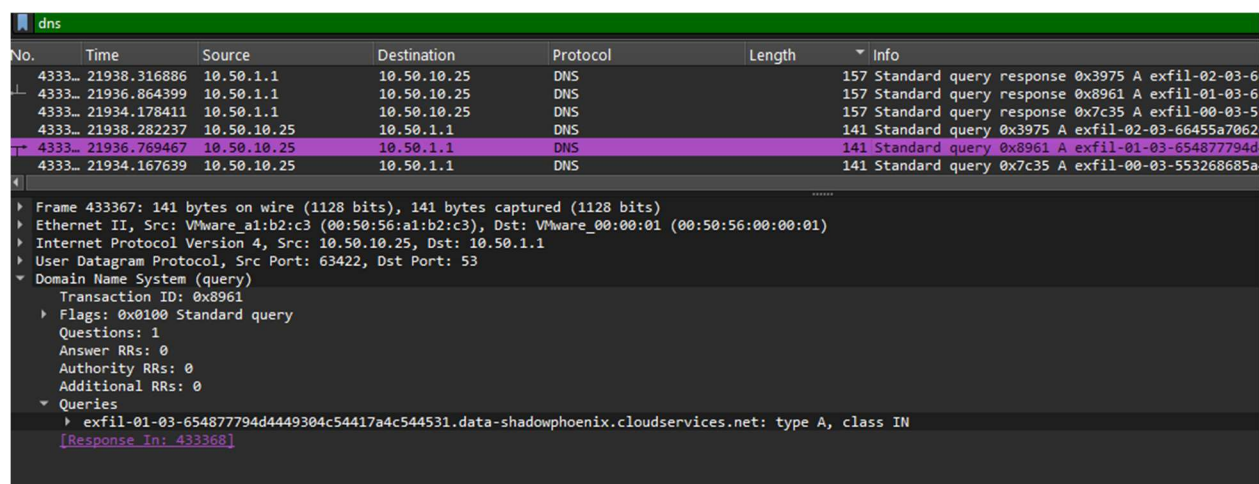


Figure 22. Extracting the Second Data Fragment from the Suspicious DNS Queries.

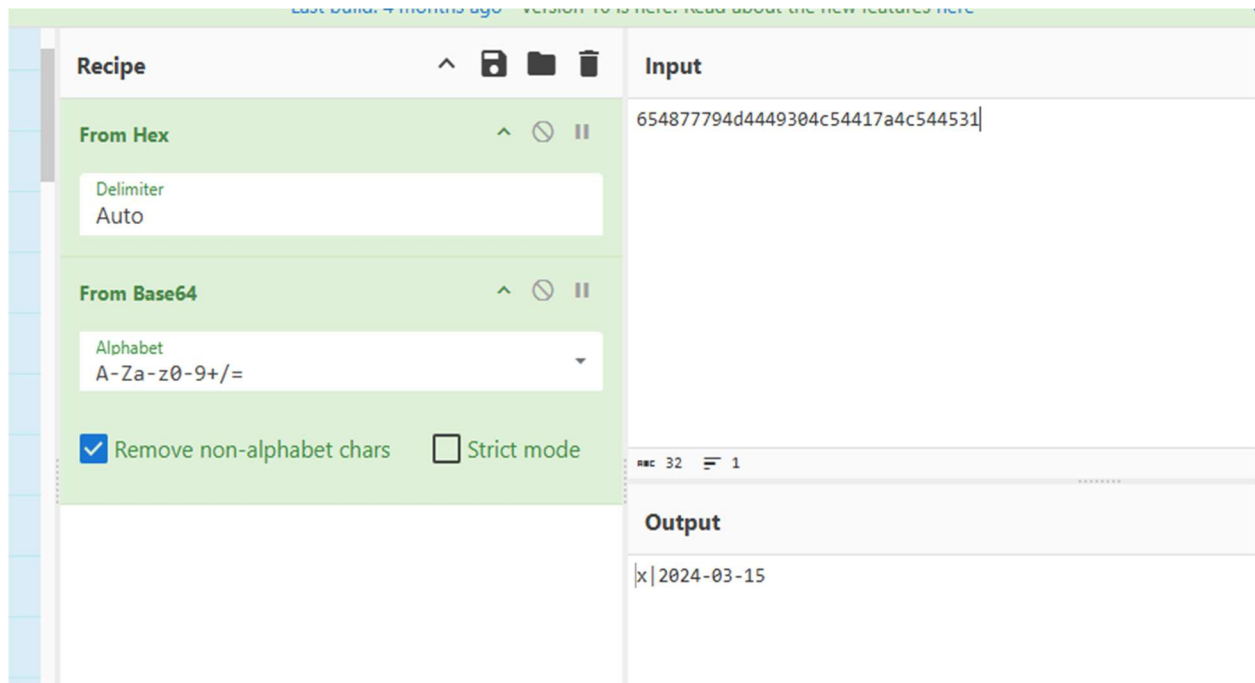


Figure 23. Decrypted Output Revealing the Exfiltration Date (2024-03-15).

Replicating the established decoding methodology (Hexadecimal → Base64) yielded the second plaintext segment: 'x | 2024-03-15 ' (see Figure 23). This artifact provided two critical insights:

1. **Attribution:** The initial character ('x') concatenates with the previous segment to complete the threat actor's name ('Shadow Phoenix').
2. **Timeline:** The string explicitly defines the **date of exfiltration** as March 15, 2024.

The investigation concluded with the analysis of the third and final payload segment:

dns						
No.	Time	Source	Destination	Protocol	Length	Info
4333...	21938.316886	10.50.1.1	10.50.10.25	DNS	157	Standard query response 0x3975 A exfil-02-03-6645
4333...	21936.864399	10.50.1.1	10.50.10.25	DNS	157	Standard query response 0x8961 A exfil-01-03-6548
4333...	21934.178411	10.50.1.1	10.50.10.25	DNS	157	Standard query response 0x7c35 A exfil-00-03-5532
4333...	21938.282237	10.50.10.25	10.50.1.1	DNS	141	Standard query 0x3975 A exfil-02-03-66455a70626d4
4333...	21936.769467	10.50.10.25	10.50.1.1	DNS	141	Standard query 0x8961 A exfil-01-03-654877794d444
4333...	21934.167639	10.50.10.25	10.50.1.1	DNS	141	Standard query 0x7c35 A exfil-00-03-553268685a473

▶ Frame 433369: 141 bytes on wire (1128 bits), 141 bytes captured (1128 bits) ▶ Ethernet II, Src: VMware_a1:b2:c3 (00:50:56:a1:b2:c3), Dst: VMware_00:00:01 (00:50:56:00:00:01) ▶ Internet Protocol Version 4, Src: 10.50.10.25, Dst: 10.50.1.1 ▶ User Datagram Protocol, Src Port: 63423, Dst Port: 53 ▶ Domain Name System (query) Transaction ID: 0x3975 Flags: 0x0100 Standard query Questions: 1 Answer RRs: 0 Authority RRs: 0 Additional RRs: 0 Queries ▶ exfil-02-03-66455a70626d46755932564551673d3d.data-shadowphoenix.cloudservices.net: type A, class IN [Response In: 433370]

Figure 24. Extracting the Last Remaining Hex Value for Analysis.

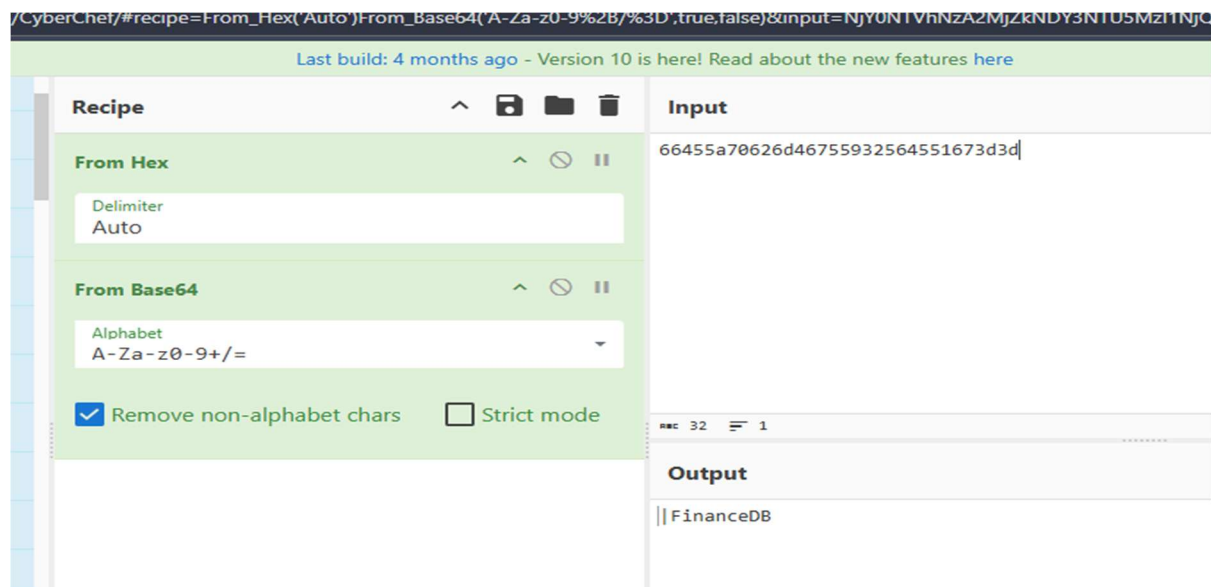


Figure 25. Final Decrypted Output Revealing the Target System (FinanceDB).

Processing the final query produced the text segment '| FinanceDB' (see Figure 25), identifying the target system. This finding critically identified the compromised host as the organization's financial database. Following this recovery, we aggregated the three decoded outputs. The reconstruction of these segments exposed the complete intelligence artifact concealed within the attacker's DNS traffic.

3.9 Assembling and Submitting the Final Flag

Following the sequential decoding and concatenation of the three DNS payloads, the complete exfiltrated message was successfully reconstructed:

ShadowPhoenix | 2024-03-15 | FinanceDB

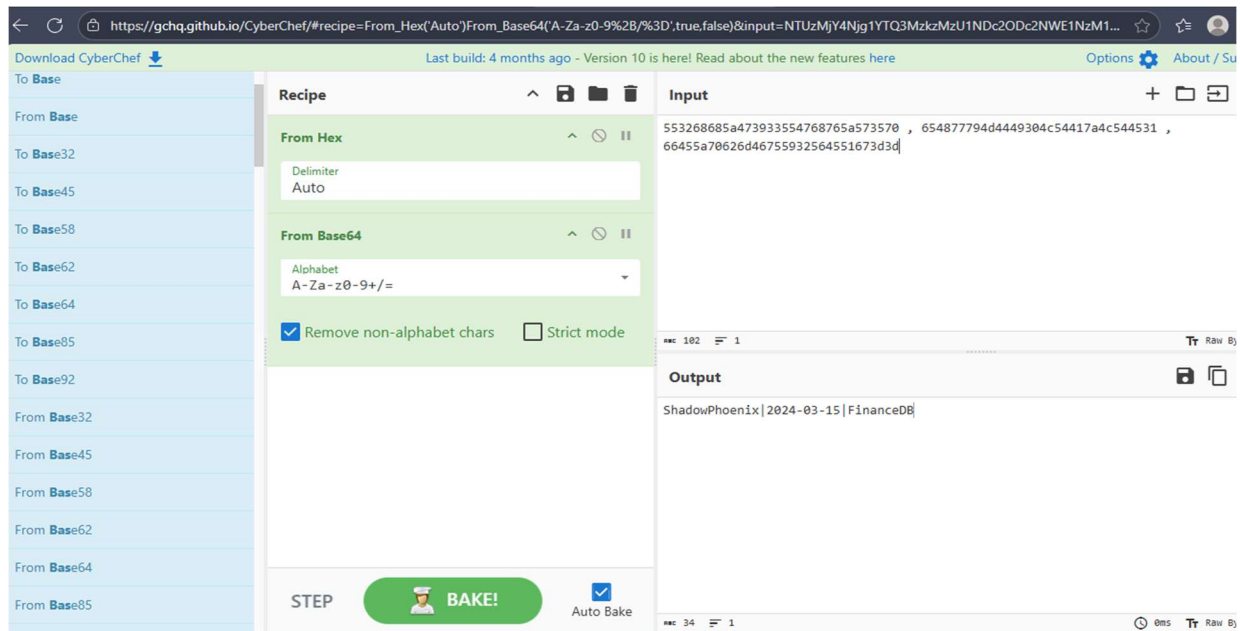


Figure 26. Hex strings decoded to reveal the APT Group, Exfiltration Date, and Target System

The cumulative forensic analysis definitively established the following intelligence parameters:

- **APT Group:** ShadowPhoenix
- **Exfiltration Date:** 2024-03-15
- **Target System:** FinanceDB

To conclude the engagement, these artifacts were synthesized in accordance with the specified flag syntax to generate the final submission:

Exploit3rs{APTGroupName_Exfil_DATE_Target-SYSTEM}

By integrating the verified intelligence artifacts into the prescribed challenge syntax, we successfully constructed the final submission string. The synthesis of the attributed APT group, exfiltration timeline, and target system yielded the conclusive flag:

Final Flag:

Exploit3rs{APTShadowPhoenix_Exfil_2024-03-15_Target-FinanceDB}

4. Problems We Ran Into

4.1 Chasing Ghosts in the Stream

Initially, we prioritized the inspection of TCP traffic, suspecting the use of steganography or data embedding within TCP payloads. This phase involved extensive stream reassembly and header analysis. While this avenue ultimately yielded negative results, it was a necessary forensic step to rule out standard exfiltration channels before pivoting to alternative protocols.

No.	Time	Source	Destination	Protocol	Length	Info
3376.	2816.035674	10.50.10.45	202.104.72.239	TCP	54	60872 → 443 [SYN] Seq=0 Win=65535 Len=0
3376.	2816.059715	202.104.72.239	10.50.10.45	TCP	54	443 → 60872 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0
3376.	2816.124408	10.50.10.45	202.104.72.239	TCP	54	60872 → 443 [ACK] Seq=1 Ack=1 Win=65535 Len=0
3376.	2816.202392	10.50.10.45	202.104.72.239	TCP	54	60872 → 443 [FIN, ACK] Seq=1 Ack=1 Win=65535 Len=0
3377.	2816.060854	10.50.10.50	116.225.26.201	TCP	54	60874 → 80 [SYN] Seq=0 Win=65535 Len=0
3377.	2816.063085	116.225.26.201	10.50.10.50	TCP	54	80 → 60874 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0
3377.	2816.145661	10.50.10.50	116.225.26.201	TCP	54	60874 → 80 [ACK] Seq=1 Ack=1 Win=65535 Len=0
3377.	2816.239449	10.50.10.50	116.225.26.201	TCP	54	60874 → 80 [FIN, ACK] Seq=1 Ack=1 Win=65535 Len=0
3377.	2816.097858	10.50.10.45	59.51.12.220	TCP	54	60876 → 443 [SYN] Seq=0 Win=65535 Len=0
3377.	2816.112638	59.51.12.220	10.50.10.45	TCP	54	443 → 60876 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0
3377.	2816.173604	10.50.10.45	59.51.12.220	TCP	54	60876 → 443 [ACK] Seq=1 Ack=1 Win=65535 Len=0
3377.	2816.234446	10.50.10.45	59.51.12.220	TCP	54	60876 → 443 [FIN, ACK] Seq=1 Ack=1 Win=65535 Len=0
3377.	2816.173934	10.50.10.45	202.13.93.81	TCP	54	60879 → 80 [SYN] Seq=0 Win=65535 Len=0
3377.	2816.231336	202.13.93.81	10.50.10.45	TCP	54	80 → 60879 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0
3377.	2816.297341	10.50.10.45	202.13.93.81	TCP	54	60879 → 80 [ACK] Seq=1 Ack=1 Win=65535 Len=0
3377.	2816.375850	10.50.10.45	202.13.93.81	TCP	54	60879 → 80 [FIN, ACK] Seq=1 Ack=1 Win=65535 Len=0
3377.	2816.205532	10.50.40.15	22.134.110.38	TCP	54	60881 → 443 [SYN] Seq=0 Win=65535 Len=0
3377.	2816.279259	22.134.110.38	10.50.40.15	TCP	54	443 → 60881 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0
3377.	2816.313189	10.50.40.15	22.134.110.38	TCP	54	60881 → 443 [ACK] Seq=1 Ack=1 Win=65535 Len=0
3377.	2816.323915	10.50.40.15	22.134.110.38	TCP	54	60881 → 443 [FIN, ACK] Seq=1 Ack=1 Win=65535 Len=0
3377.	2816.292675	10.50.10.50	221.143.138.193	TCP	54	60883 → 443 [SYN] Seq=0 Win=65535 Len=0
3377.	2816.355276	221.143.138.193	10.50.10.50	TCP	54	443 → 60883 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0

Figure 27. Network Traffic Filtered to Display Only TCP Packets.

However, detailed inspection of the TCP streams revealed a consistent pattern of zero-payload connections. The traffic was strictly limited to standard handshake and termination sequences (see Figure 27, Info column), without any accompanying data transfer or anomalous artifacts. This comprehensive lack of actionable intelligence indicated that the TCP layer was likely benign background noise rather than a vector for exfiltration.

Strategic Pivot:

Acknowledging the null results from the TCP analysis, we concluded that further scrutiny of this protocol would yield diminishing returns. Consequently, we redirected our investigative focus to the Simple Mail Transfer Protocol (SMTP). This methodological adjustment proved decisive, immediately exposing the initial Indicators of Compromise (IoC).

4.2 The False Lead: Export Objects

Another challenge we encountered was the time spent reviewing the Export Objects section in Wireshark. We went through each category one by one, including HTTP, SMB, TFTP, IMF, and FTP-DATA (see Figure 28). Our assumption was that the attacker might have left something behind or embedded a file within one of these exported objects, so we examined each option with the expectation of uncovering a hidden artifact.

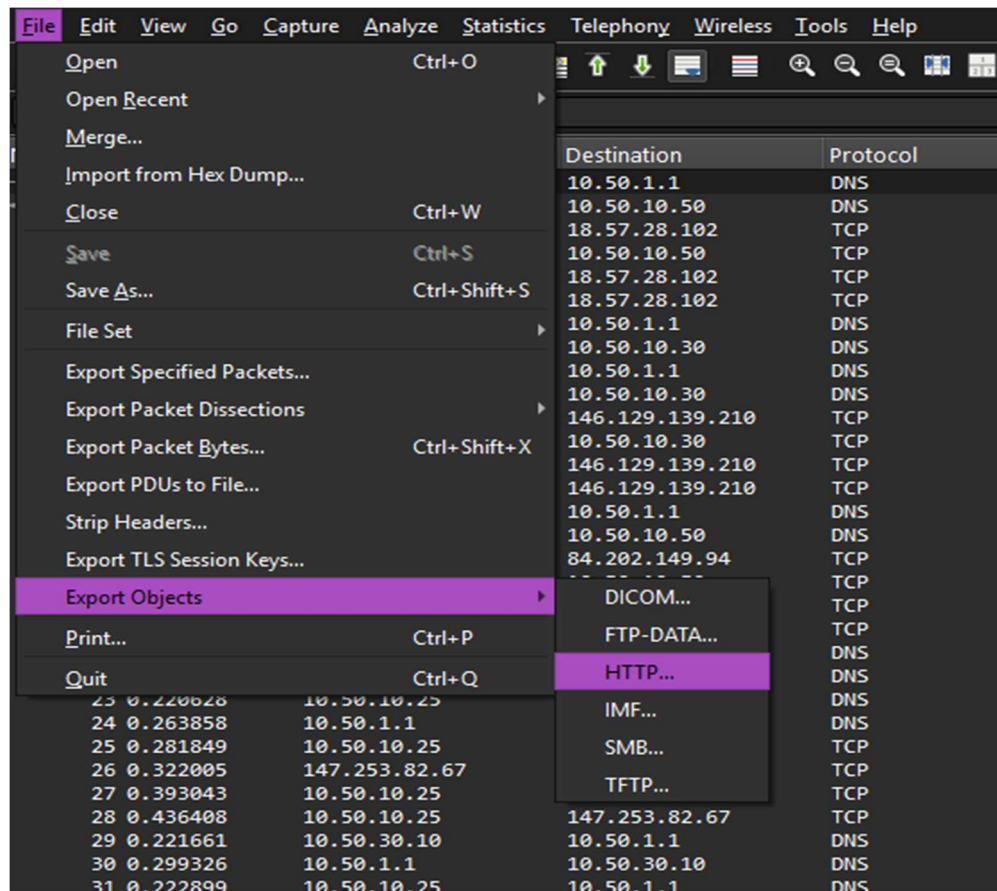


Figure 28. Navigating to File > Export Objects > HTTP in Wireshark

However, all of these sections were empty, with no objects or relevant data. Each category we examined returned zero results, and even after revisiting multiple options, the outcome remained unchanged.

Retrospective Analysis: These protocols were not used in the challenge. The attacker did not rely on any of them, which meant that reviewing the Export Objects menu ultimately contributed no value to the investigation and resulted in unnecessary effort.

5. Technical Environment and Tools

5.1 ChatGPT

Throughout the engagement, ChatGPT served as a reference interface to augment the team's domain knowledge. Facing complex indicators related to packet analysis and DNS tunneling, we leveraged the tool to clarify conceptual frameworks (see Figure 29) and interpret the anomalous data structures extracted from the capture file (see Figure 30). This ensured accurate identification of the obfuscation methods employed by the threat actor.

1. Overall Network Activity & Background Traffic — Key Concepts

What you should understand / accept from this kind of PCAP

- A PCAP usually contains **both normal traffic and potential malicious traffic mixed together**.
- Normal browsing traffic like DNS requests, HTTP requests, or CDN activity creates **background noise**.
- Attackers often hide malicious actions inside this noise so it looks like normal network usage.

Figure 29. Key concepts regarding PCAP analysis

What this means for you when investigating

- You must expect **a lot of packets** that are irrelevant to the attack.
- The challenge is to **separate signal from noise**, meaning:
 - Identify which traffic is normal.
 - Identify which traffic does not fit the pattern.

Figure 30. Guidelines for filtering out irrelevant packets to isolate malicious traffic patterns.

5.2 Wireshark



Figure 31 Wireshark Logo

The primary instrument employed for this engagement was Wireshark, an industry-standard network protocol analyzer (CTF101, 2024). This software facilitated the ingestion of the provided PCAP artifact and enabled granular, packet-level inspection. Such deep analysis was prerequisite for reconstructing the adversary's exfiltration methodology. Given the challenge's emphasis on anomaly detection within network streams, Wireshark served as the central pillar of our forensic workflow.

Why Wireshark?

- **Deep Packet Inspection:** Wireshark provides visibility into every protocol layer, including Ethernet, IP, TCP or UDP, DNS, and SMTP. This allowed us to examine each suspicious packet in detail and understand the exact data being transmitted.
- **Advanced filtering capabilities:** Its display filters enabled us to isolate specific traffic types, such as DNS, SMTP, or TCP flows, which was essential given the significant amount of background noise in the PCAP.
- **Protocol Analysis:** The challenge required close examination of DNS queries, particularly those containing encoded data. Wireshark's protocol decoding and information panels helped us distinguish abnormal DNS behavior from legitimate traffic.
- **Stream Reconstruction:** Tools like Follow TCP Stream and Follow UDP Stream allowed us to view entire communication flows rather than isolated packets. This was crucial for analyzing the complete SMTP email containing the malicious macro and for tracking the DNS requests used in the exfiltration process.

5.3 CyberChef



Figure 32. CyberChef Logo

Following the extraction of anomalous DNS payloads from Wireshark, CyberChef was utilized as the primary environment for artifact reconstruction. Given the evident use of data obfuscation by the threat actor, the investigation required a flexible framework to conduct iterative cryptanalysis. CyberChef's modular architecture facilitated the rapid prototyping of decoding pipelines, enabling the dynamic chaining of operations, and providing real-time validation of the resulting plaintext.

Why CyberChef?

- **Efficient decoding workflow:** With operations such as *From Hex* and *From Base64*, CyberChef enabled systematic testing of multiple decoding methods without the need for scripting.
- **Layered analysis capabilities:** When initial outputs were not interpretable, CyberChef allowed us to apply additional operations sequentially, reflecting the attacker's use of multi-layer encoding.
- **Immediate feedback:** The tool updated results in real time as adjustments were made, allowing us to quickly identify which decoding steps were effective.

CyberChef functioned as the primary transformation engine, converting raw artifacts extracted via Wireshark into actionable intelligence. By abstracting the complexity of cryptographic operations, it served as the critical bridge between artifact extraction and final payload reconstruction, effectively synthesizing the disparate data points into a cohesive forensic narrative.

6. REFLECTIONS AND LESSONS LEARNT

6.1 Technical Skills Gained

Throughout the Operation Shadow Phoenix challenge, our team developed a deeper understanding of network traffic analysis, particularly through the use of Wireshark. This challenge required us to analyze packet captures with precision, identify abnormal communication patterns, and trace the flow of data to uncover hidden indicators within the traffic. By working through multiple iterations of filtering, decoding protocols, and interpreting packet structures, we strengthened our ability to analyze real-world network artifacts systematically. This experience not only improved our familiarity with Wireshark's toolset but also enhanced our confidence in tackling network-centric forensics under time-sensitive conditions.

Beyond the primary challenge, the competition also exposed us to a wide range of other technical domains. In Nether Network Mystery (Reverse Engineering), we practiced navigating binary logic, understanding compiled code behavior, and applying basic static and dynamic analysis techniques. Meanwhile, Digital Fortress (PWN) pushed us to recognize memory vulnerabilities and reason about how insecure program logic can be weaponized. Even though these challenges were not selected for the full writeup, they served as valuable opportunities to broaden our baseline competencies across different CTF categories.

6.2 Challenges Faced and How We Overcame Them

One of the earliest challenges we encountered was **time management**, especially during the initial rush to secure and solve high-value tasks. Most of us focused on the same challenges, which unintentionally slowed down our overall progress. Once we recognized the inefficiency, we reorganized ourselves by dividing the workload and assigning different challenges to different members. This simple shift allowed us to progress in parallel and greatly improved the team's momentum.

On top of that, we had to manage time constraints and pressure due to **personal issues** faced by one of our teammates, Taimoor, who could only participate at the beginning of the System Override competition. His absence added some pressure, but we adapted quickly and redistributed the workload to ensure steady progress. Despite the setback, we remained determined and continued giving our best as a team.

Lastly, the **difficulty level of the competition** introduced a fair amount of stress, especially since the challenges were more complex than what we had practiced in class. This made it easy to feel stuck or underprepared. To overcome this, we communicated openly, shared resources, and made use of external references such as web documentation and AI tools whenever appropriate. Through collaboration and adaptability, we gradually found our footing and pushed through the more demanding tasks.

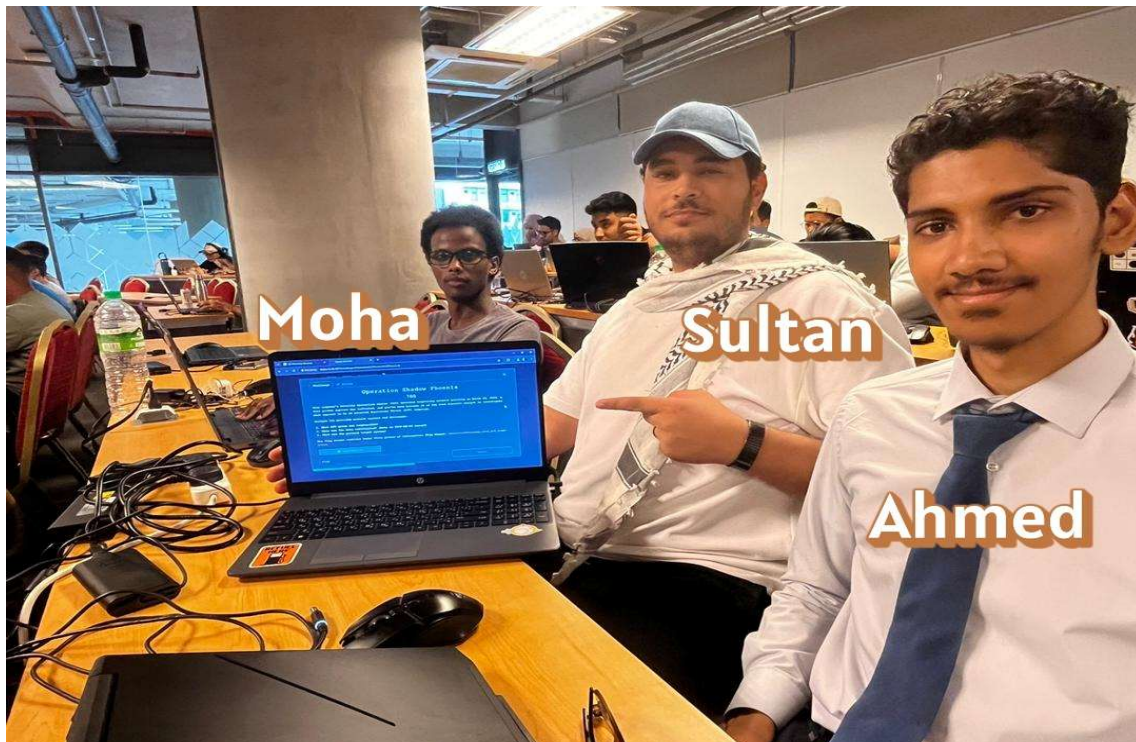
6.3 What Would We Do Differently

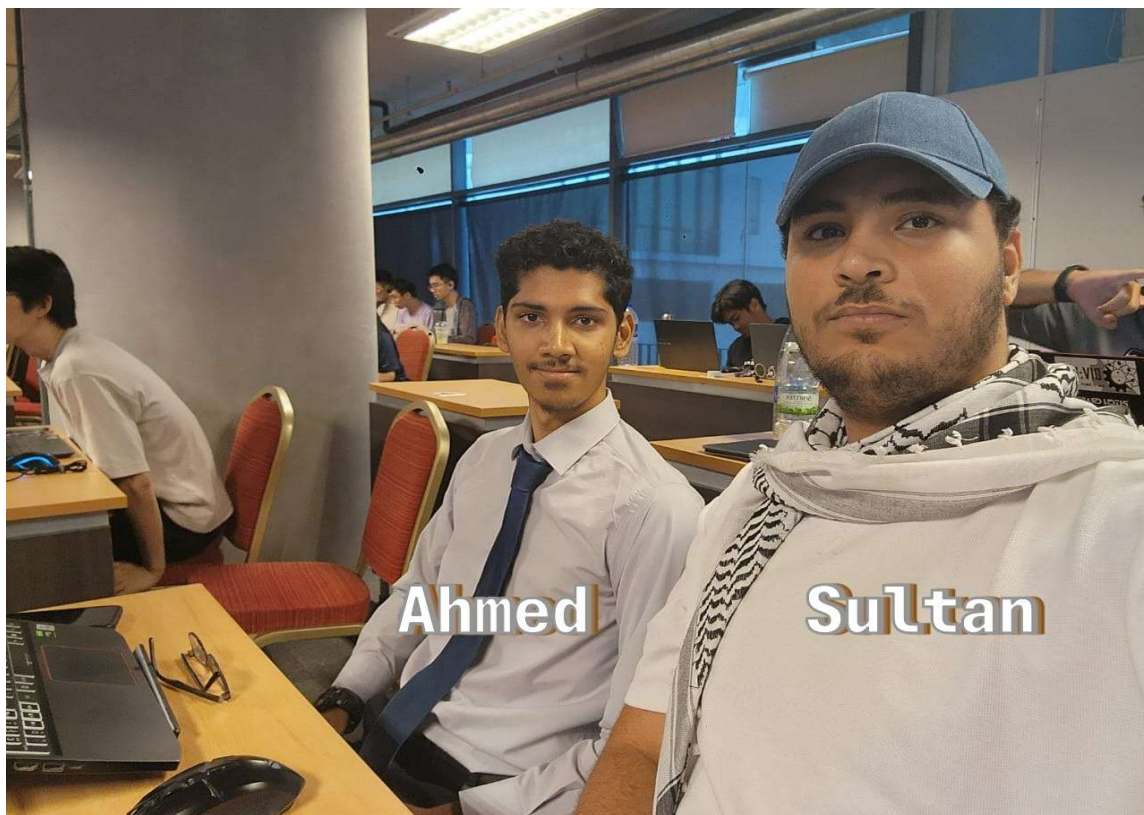
After competing in the System Override CTF and dealing with a few setbacks, we realized we need to sharpen our strategies if we want stronger results next time. For example:

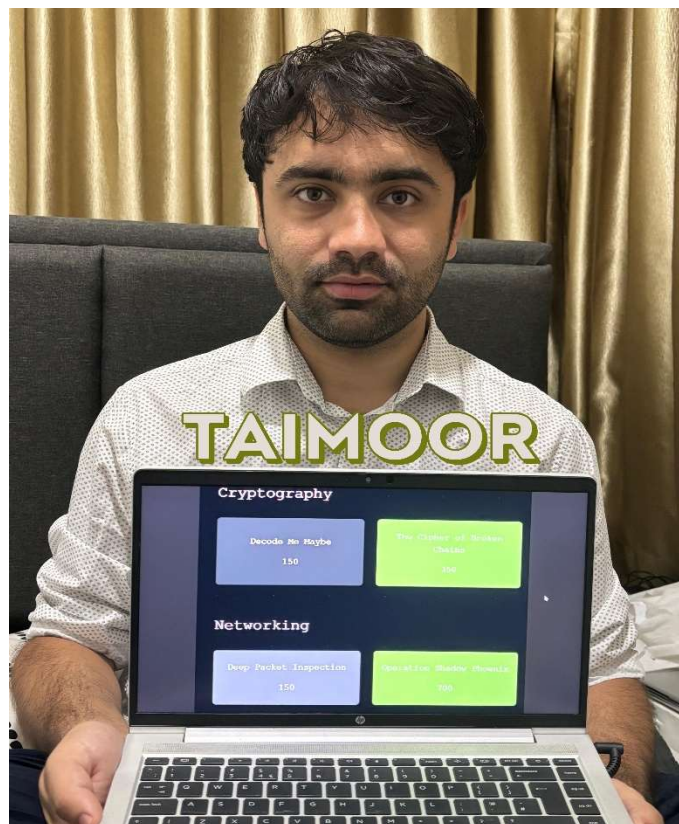
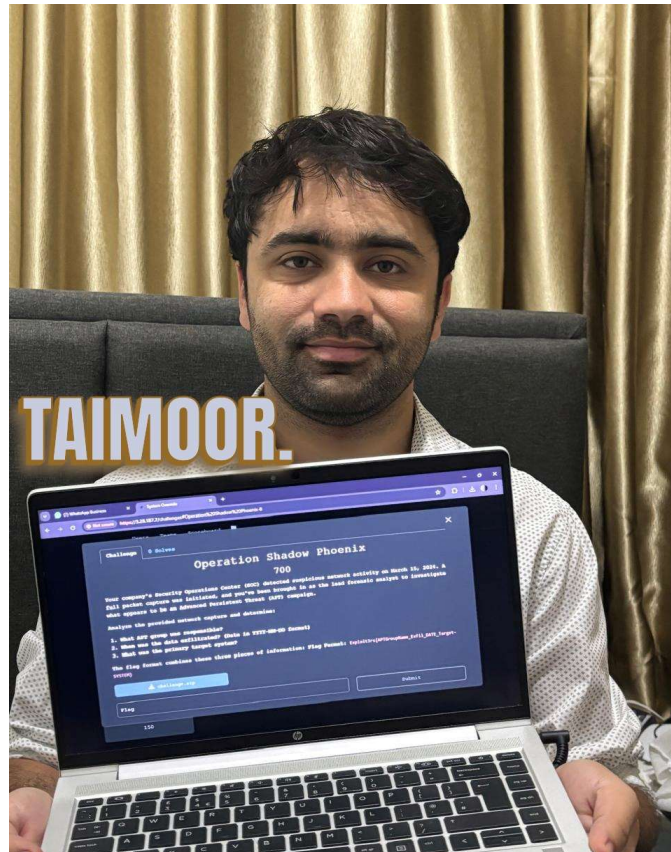
- We would start by researching and practicing the key topics of the competition well in advance, especially networking, cryptography, and reverse engineering, so that we enter the event with stronger fundamentals and greater confidence. Early preparation would help us solve challenges more efficiently under pressure.
- We would also create a structured task-distribution plan from the beginning, assigning each team member to different categories instead of clustering around the same challenge. This would prevent time wastage, reduce overlap, and allow us to progress in parallel rather than sequentially.
- Another improvement would be to establish clearer decision-making rules, such as setting strict time limits for difficult challenges and rotating tasks more systematically. This would help prevent situations where conflicting ideas slow us down or where we remain stuck for too long.
- Lastly, we would allocate backup strategies for unexpected situations, such as a team member being unable to participate fully. Having predefined contingency plans and flexible workload distribution would help us remain steady even when facing unforeseen constraints.

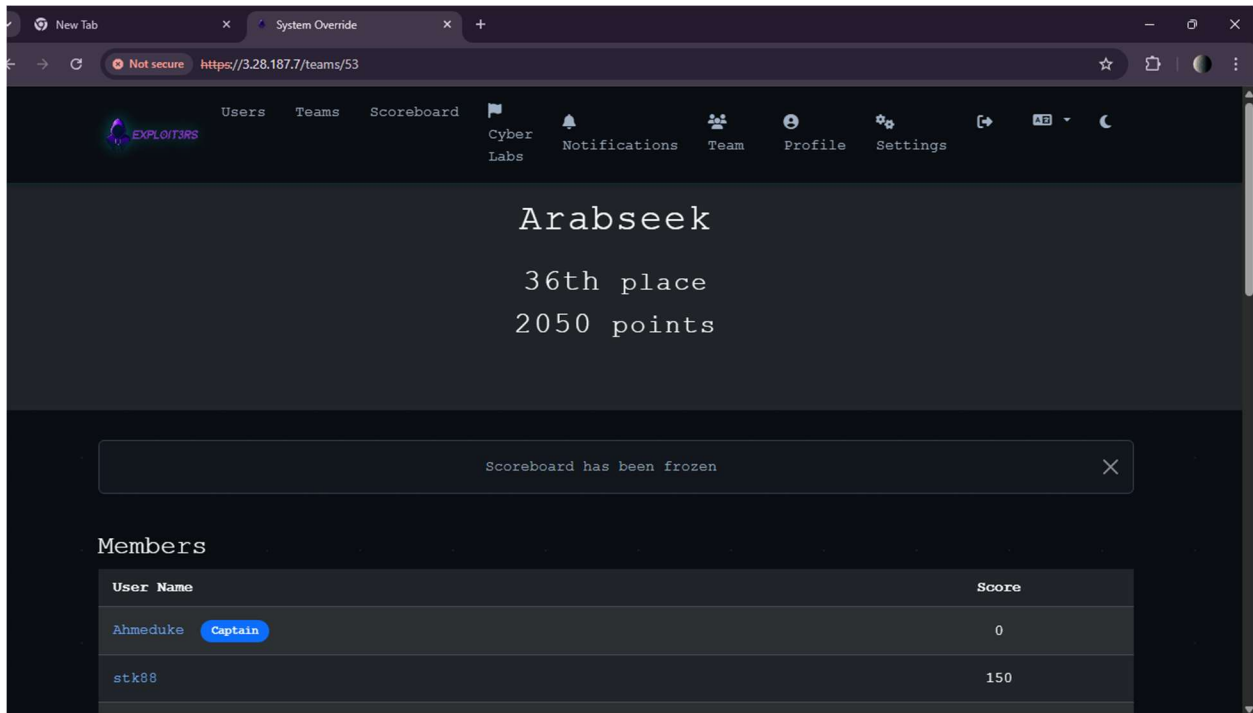
6.4 Proof of Participation in CTF Competitions

6.4.1 System Override CTF









EXPLOIT3RS

Users Teams Scoreboard Cyber Labs Notifications Team Profile Settings

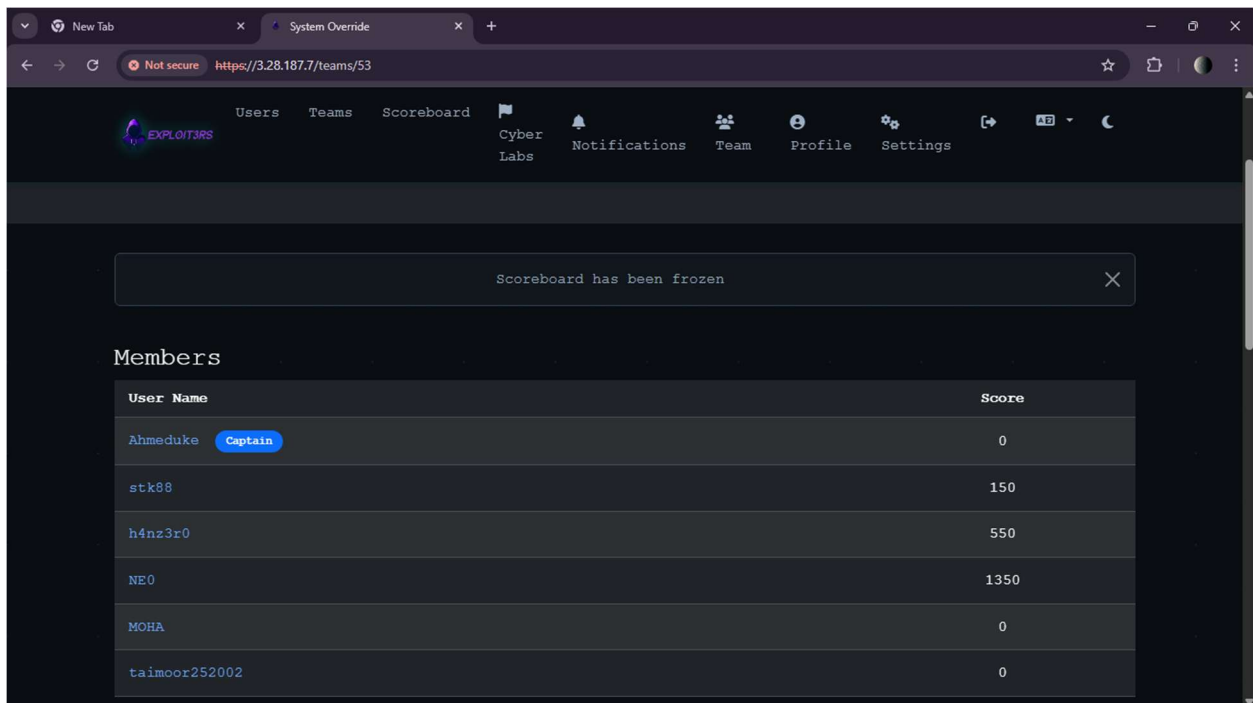
Arabseek

36th place
2050 points

Scoreboard has been frozen

Members

User Name	Score
Ahmeduke Captain	0
stk88	150



EXPLOIT3RS

Users Teams Scoreboard Cyber Labs Notifications Team Profile Settings

Arabseek

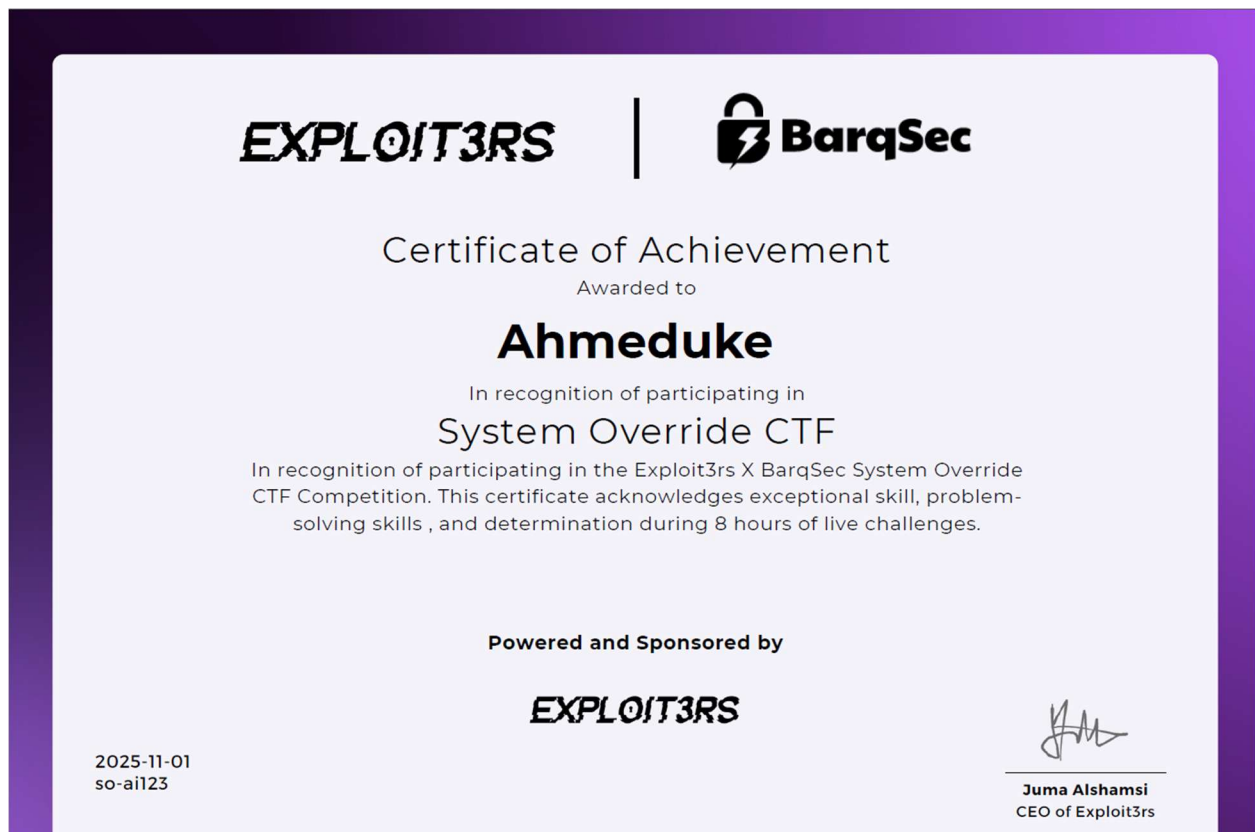
36th place
2050 points

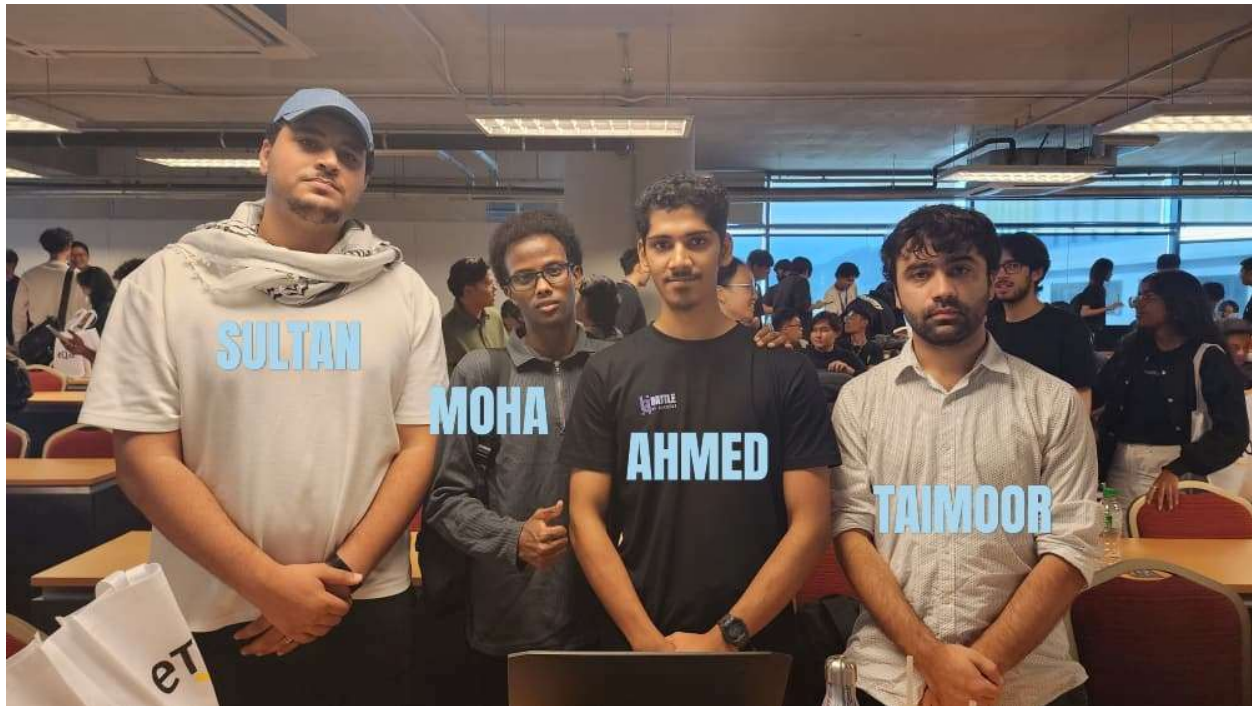
Scoreboard has been frozen

Members

User Name	Score
Ahmeduke Captain	0
stk88	150
h4nz3r0	550
NE0	1350
MOHA	0
taimoor252002	0





6.4.2 International Battle of Hackers (IBOH)





6.4.3 Bingo CTF







BINGOCTF Challenges Notifications Team Profile Settings

Members

User Name	Score
AHMED MIRAHUSAIN ALVI Captain	
stk00	
Muhammad Taimoor Jamal	

Solves

Challenge	Category	Value	Time
Rock Paper Scissors	Web	235	November 29th, 11:46:52 AM

● Solves (50.00%) ● Fails (50.00%) ● Web (100%)

BINGOCTF Challenges Notifications Team Profile Settings

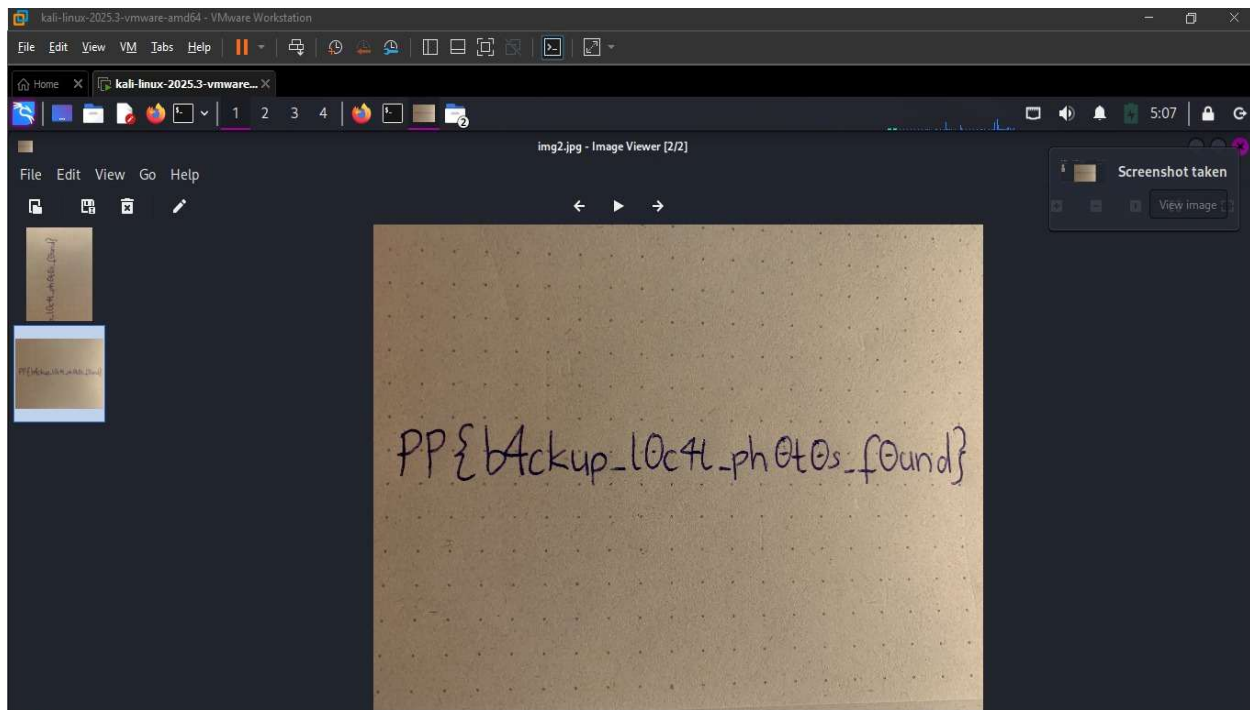
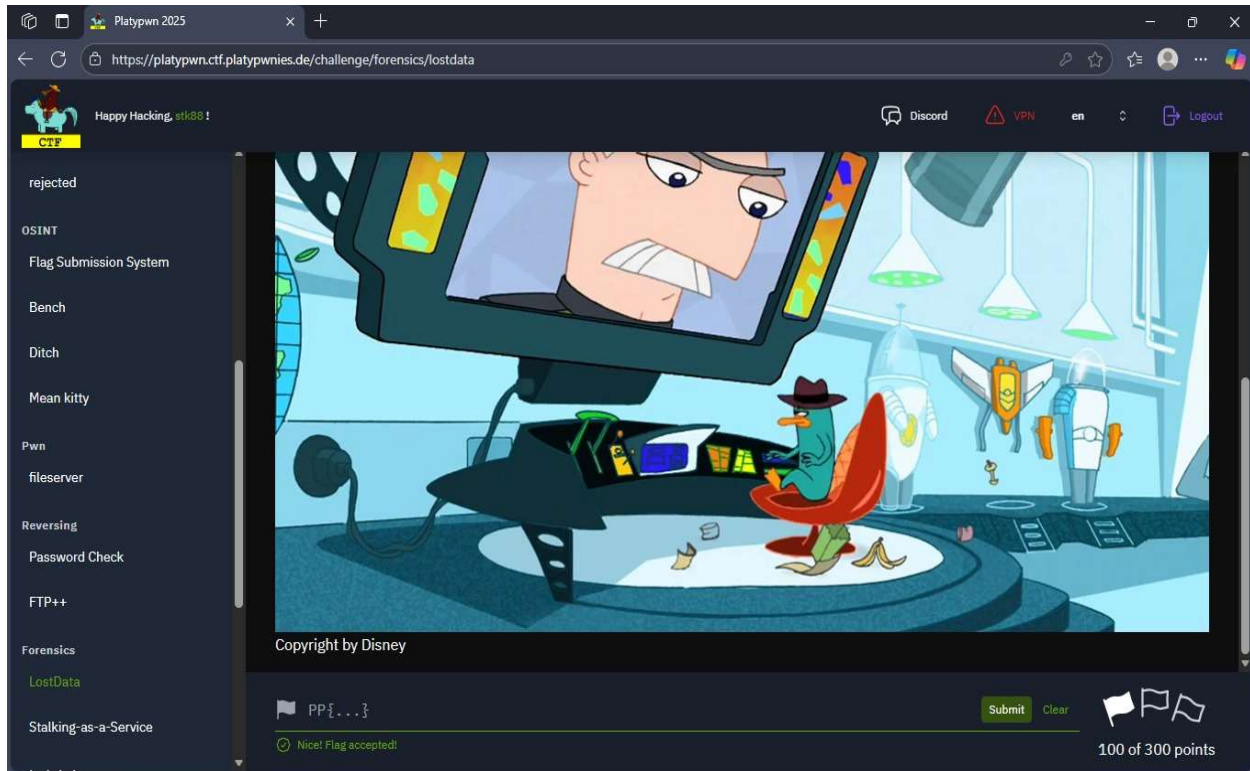
MOHA

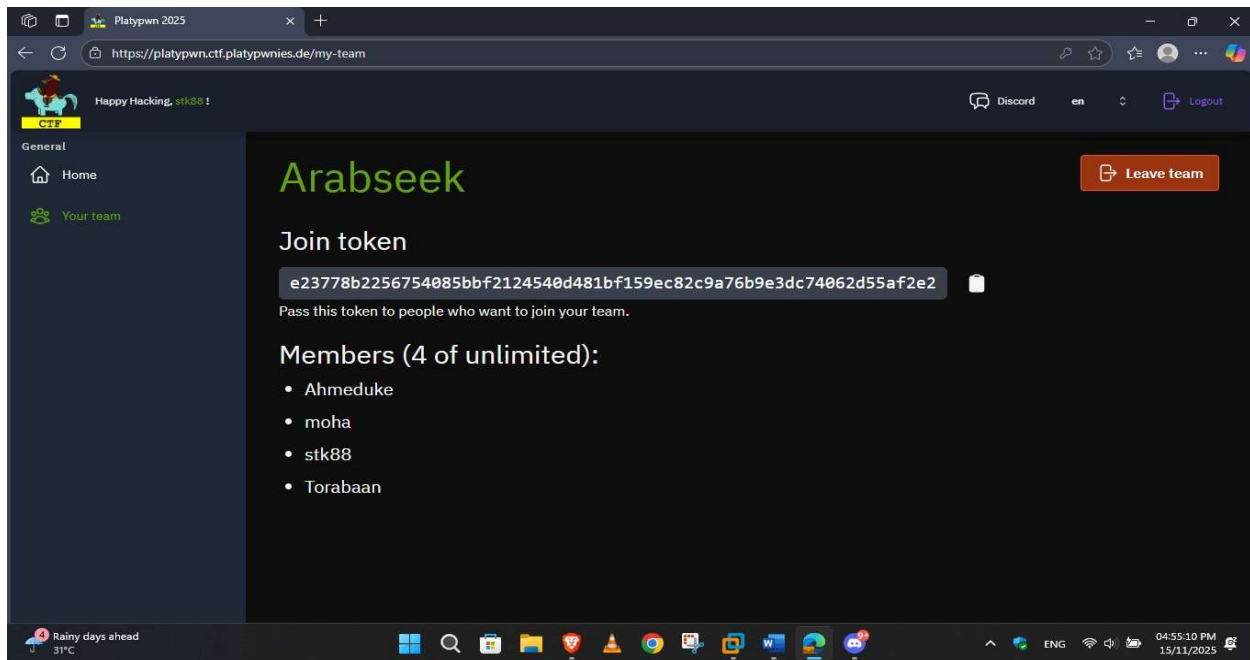
⚙️ 👤 📄 🗑️

Members

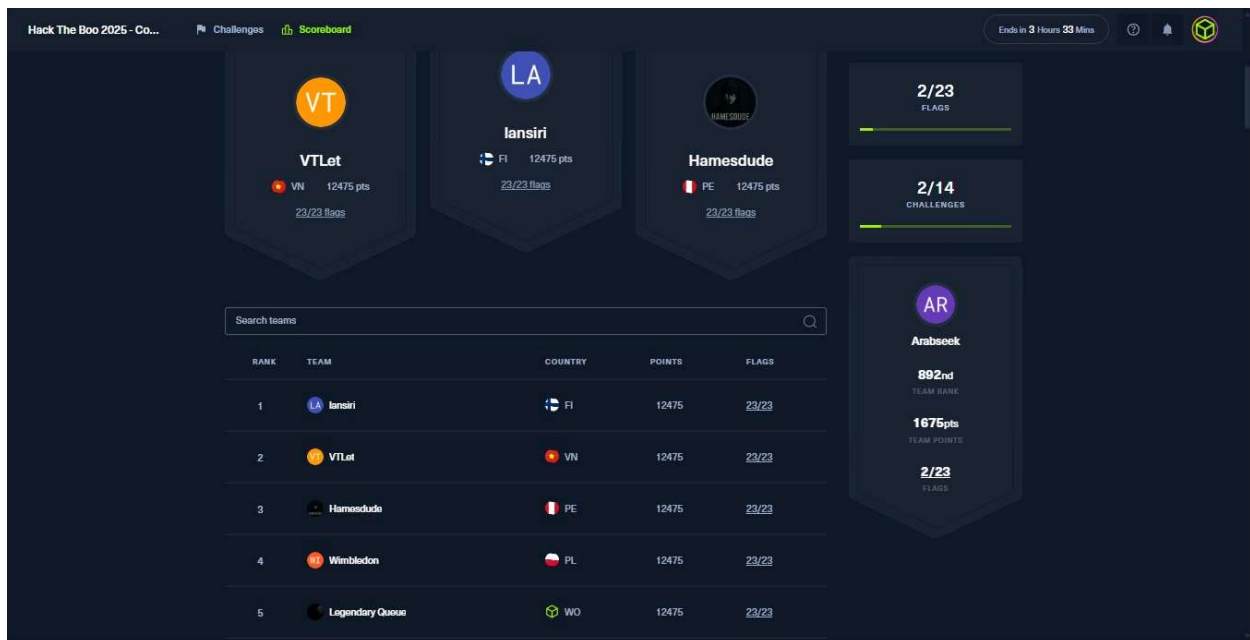
User Name	Score
MOHA Captain	

6.4.4 PlatyPwn CTF (Online)





6.4.5 HackTheBoo 2025 – Competition (Online)



Aside from the System Override CTF, which was the main tournament for our assignment, we also joined several other competitions. These included IBOH 2025 and Bingo CTF, both hosted by the FSEC-SS club at APU, along with PlatyPwn CTF and the HackTheBoo 2025 competition.

Our team solved many of the CTF challenges we attempted, but in the end, we selected the System Override CTF as our final competition for the group assignment.

7. CONCLUSION

7.1 Summary of Achievements

During the System Override 2025 CTF, our team successfully completed several challenges across multiple categories, with particular emphasis on networking through the Operation *Shadow Phoenix* challenge. We demonstrated strong teamwork by adapting quickly to time constraints, distributing tasks effectively, and maintaining consistent communication throughout the event. Our ability to apply analytical techniques such as packet capture analysis, reverse engineering fundamentals, and vulnerability identification highlighted our growing technical proficiency. Most importantly, the competition enabled us to significantly enhance our problem-solving capabilities, develop resilience under pressure, and strengthen our collaborative workflow, marking a meaningful step forward in our collective cybersecurity learning journey.

7.2 Importance of CTFs in Cybersecurity Education

Capture The Flag (CTF) challenges have become integral to cybersecurity education, offering hands-on, practical experiences that bridge the gap between theoretical knowledge and real-world applications. Unlike traditional classroom learning, CTF exercises immerse participants in realistic security scenarios where they must identify vulnerabilities and exploit systems in a controlled, ethical manner, developing critical thinking skills and creative problem-solving abilities.

The educational value of CTFs lies in their breadth and depth. Participants gain exposure to multiple cybersecurity domains—including web exploitation, reverse engineering, cryptography, digital forensics, and steganography—becoming more well-rounded security professionals. This multi-disciplinary approach ensures learners understand how different attack vectors and defense mechanisms work in concert.

Many employers actively seek CTF experience, viewing it as evidence of practical competence. Additionally, the competitive yet fun nature of CTFs heightens engagement and motivation, making cybersecurity an attractive career path for newcomers to the field. (Tidmarsh, 2025)

7.3 Future Aspirations

Participation in the System Override 2025 CTF significantly reinforced our theoretical understanding of cybersecurity principles by applying them in a simulated adversarial environment. Through challenges necessitating skills in networking, reverse engineering, cryptography, and exploitation, the team bridged the gap between academic concepts and practical execution. These experiences highlighted the necessity of adaptability under pressure and have motivated a strategic focus on offensive security methodologies, specifically within penetration testing and red teaming.

Looking ahead, we intend to participate in more CTF competitions to broaden our technical capabilities and improve our teamwork. Engaging with the wider cybersecurity community will allow us to connect with experienced professionals, learn new methodologies, and increase our chances of securing internships or future job opportunities. CTF events often attract skilled players and industry recruiters, making them an excellent platform for both growth and career development.

Lastly, we are considering pathways in cybersecurity engineering, incident response, or threat intelligence, where we can apply our analytical and defensive skills while still tackling complex challenges. Our long-term goal is to stay active in the CTF community, contribute to its growth, and pass on the knowledge we gain to future participants.

8. WORKLOAD MATRIX

No	TP	Name	Component
1	TP085327	SULTAN ABDULLA OMAR TAKRORI	❖ Technical Execution: Spearheaded the development of the comprehensive, CTF Writeup. ❖ Problem Solving: Successfully resolved the majority of the technical obstacles within the challenge.
2	TP082459	MOHAMED ABDIFATAH ALI	❖ Investigative Support: Provided technical assistance throughout the challenge execution. ❖ Report Structure: Authored the Conclusion section and organized the Table of Contents and Table of Figures.
3	TP084807	AHMED MIRAHUSAIN ALVI	❖ Requirement Analysis: Facilitated the team's initial understanding of the challenge scope and requirements. ❖ Documentation: Contributed to the drafting of the technical documentation.
4	TP085061	MUHAMMAD TAIMOOR JAMAL	❖ Artifact Analysis: Executed specific technical tasks related to decoding strings. ❖ Report Compilation: Managed the final documentation process and ensured proper formatting of the report.

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